

# **Post Stroke Speech Therapy for Adults**



**Muhammad Abdur Rehman (Rc-449)**

**Ahmad Waseem (Rc-460)**

**Muhammad Bilal (Rc-441)**

*Supervised By*

**Sir Imran Javed**

*Co-Supervised By*

**Sir Ahmad**

*Submitted for the partial fulfillment of BS Computer  
Science*

**Department of Computer Science**

**NUML, Rawalpindi**

**May, 2026**

# **ABSTRACT**

Following a stroke, speech difficulties can seriously affect a person's capacity to communicate, resulting in mental suffering and social isolation. Many individuals who have trouble speaking after a stroke are ashamed to go to a professional therapist in person.

Our Final Year Project (FYP) aims to address this issue by developing a mobile application that provides AI-driven speech therapy for adults recovering from stroke. This app will feature interactive exercises, personalized therapy plans, real-time speech recognition feedback, and gamification elements to help users regain their speech abilities in the comfort and privacy of their homes.

The app will have AI-driven voice analysis that uses NLP models and the Google voice-to-Text API to enable real-time pronunciation evaluation. Additionally, it will provide customized therapy sessions according to each user's development using adaptive learning strategies. The app will also include speech-based interactive activities that are intended to enhance articulation and fluency in a fun and interesting way.

This project aims to reduce the stigma associated with speech therapy by providing a self-paced, accessible, and affordable alternative to traditional rehabilitation.

## CERTIFICATE

Dated: \_\_\_\_\_

## FINAL APPROVAL

It is certified that project report titled „**Post Stroke Speech Therapy For Adults**“ submitted by **Muhammad Abdur Rehman, Muhammad Bilal and Ahmad Waseem** for the partial fulfillment of the requirement of “**Bachelor’s Degree in Computer Science**” is approved.

## COMMITTEE

Dr. Junaid Tariq  
HoD Computer Science:

Signature: \_\_\_\_\_

Ms. Sobia Shafiq  
Project Coordinator:

Signature: \_\_\_\_\_

Mr. Imran Javed  
Supervisor:

Signature: \_\_\_\_\_

## DECLARATION

We hereby declare that our dissertation is entirely our work and genuine / original. We understand that in case of discovery of any PLAGIARISM at any stage, our group will be assigned an F (FAIL) grade and it may result in withdrawal of our Bachelor's degree.

Group members:

**Name**

**Signature**

Muhammad Abdur Rehman

---

Muhammad Bilal

---

Ahmad Waseem

---

## PLAIGRISM CERTIFICATE

This is to certify that the project entitled “**Post Stroke Speech Therapy For Adults**”, which is being submitted here with for the award of the “**Degree of Bachelor’s/Master’s in Computer Science**”. This is the result of the original work by **Muhammad Abdur Rehman, Muhammad Bilal** and **Ahmad Waseem** under my supervision and guidance. The work embodied in this project has not been done earlier for the basis of award of any degree or compatible certificate or similar tile of this for any other diploma/examining body or university to the best of my knowledge and belief.s

### Turnitin Originality Report

Processed on 19-July-2026 06:58

ID: 30000039655

Word Count: 13681

Similarity Index

11%

Similarity by Source

|                   |     |
|-------------------|-----|
| Internet Sources: | 9%  |
| Publications:     | 4%  |
| Student Papers:   | 8 % |

---

Date: 20/07/2026

Mr. Imran Javed (Supervisor)

## FYP VOCAL

### ORIGINALITY REPORT

|                  |                  |              |                |
|------------------|------------------|--------------|----------------|
| <b>11</b> %      | <b>9</b> %       | <b>4</b> %   | <b>8</b> %     |
| SIMILARITY INDEX | INTERNET SOURCES | PUBLICATIONS | STUDENT PAPERS |

### PRIMARY SOURCES

|           |                                                                                                                                                                                                                                                                   |                |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>1</b>  | <b>Submitted to Higher Education Commission Pakistan</b><br>Student Paper                                                                                                                                                                                         | <b>3</b> %     |
| <b>2</b>  | <b>Submitted to International Institute of Science Art and Technology IISAT, Gujranwala</b><br>Student Paper                                                                                                                                                      | <b>&lt;1</b> % |
| <b>3</b>  | <b>research.superior.edu.pk</b><br>Internet Source                                                                                                                                                                                                                | <b>&lt;1</b> % |
| <b>4</b>  | <b>Submitted to Vidyalankar Institute of Technology</b><br>Student Paper                                                                                                                                                                                          | <b>&lt;1</b> % |
| <b>5</b>  | <b>www.mdpi.com</b><br>Internet Source                                                                                                                                                                                                                            | <b>&lt;1</b> % |
| <b>6</b>  | <b>Submitted to International Institute of Information Technology, Hyderabad</b><br>Student Paper                                                                                                                                                                 | <b>&lt;1</b> % |
| <b>7</b>  | <b>Submitted to info.administrator</b><br>Student Paper                                                                                                                                                                                                           | <b>&lt;1</b> % |
| <b>8</b>  | <b>Submitted to National University of Modern Languages</b><br>Student Paper                                                                                                                                                                                      | <b>&lt;1</b> % |
| <b>9</b>  | <b>dl.ucsc.cmb.ac.lk</b><br>Internet Source                                                                                                                                                                                                                       | <b>&lt;1</b> % |
| <b>10</b> | <b>Kinda Khalaf, Shibani Hamsa, Mubeena Ashraf Kizhakkethil, Sabahat Asim Wasti, Herbert F. Jelinek. "Integrating Physiological Signals With Technology-Driven Stroke Rehabilitation: A Systematic Review", IEEE Access, 2026</b><br>Publication                  | <b>&lt;1</b> % |
| <b>11</b> | <b>Ziming Yin, Yanhao Gong, Jing He, Ling Ren, Xin Li, Xianrui Hu, Yu Pan, Hongliu Yu. "Dual-Knowledge-Driven Interpretable Decision Support System for Stroke Critical-Care Rehabilitation: Design and Multi-Center Study", IEEE Access, 2025</b><br>Publication | <b>&lt;1</b> % |
| <b>12</b> | <b>doaj.org</b><br>Internet Source                                                                                                                                                                                                                                | <b>&lt;1</b> % |
| <b>13</b> | <b>Submitted to Dallas Community College District</b><br>Student Paper                                                                                                                                                                                            | <b>&lt;1</b> % |

|    |                                                                                                                                                                                                                |      |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 14 | Submitted to Holy Child Central Colleges Inc.<br>Student Paper                                                                                                                                                 | <1 % |
| 15 | Hongyu Liu, Dianguo Cao, Jinqiang Wang, Zhongcai Zhang, Yuqiang Wu. "Modeling and Control Analysis of Horizontal Cable-Driven Rehabilitation Robot", 2024 China Automation Congress (CAC), 2024<br>Publication | <1 % |
| 16 | <a href="http://www.numl.edu.pk">www.numl.edu.pk</a><br>Internet Source                                                                                                                                        | <1 % |
| 17 | <a href="http://dspace.christcollegeijk.edu.in">dspace.christcollegeijk.edu.in</a><br>Internet Source                                                                                                          | <1 % |
| 18 | Submitted to University of Shajrah<br>Student Paper                                                                                                                                                            | <1 % |
| 19 | <a href="http://pub.uni-bielefeld.de">pub.uni-bielefeld.de</a><br>Internet Source                                                                                                                              | <1 % |
| 20 | <a href="http://www.ijirset.com">www.ijirset.com</a><br>Internet Source                                                                                                                                        | <1 % |
| 21 | <a href="http://healthinformaticsjournal.com">healthinformaticsjournal.com</a><br>Internet Source                                                                                                              | <1 % |
| 22 | Barnett, Rachel G.. "A Qualitative Evaluation of Speech Sound Disorder Treatment Apps for Children", San Jose State University, 2024<br>Publication                                                            | <1 % |
| 23 | Submitted to Universiti Malaysia Sabah<br>Student Paper                                                                                                                                                        | <1 % |
| 24 | Submitted to University of Sheffield<br>Student Paper                                                                                                                                                          | <1 % |
| 25 | Submitted to University of Wales central institutions<br>Student Paper                                                                                                                                         | <1 % |
| 26 | <a href="http://eprints.utar.edu.my">eprints.utar.edu.my</a><br>Internet Source                                                                                                                                | <1 % |
| 27 | <a href="http://healthfunny.blogspot.com">healthfunny.blogspot.com</a><br>Internet Source                                                                                                                      | <1 % |
| 28 | <a href="http://www.coursehero.com">www.coursehero.com</a><br>Internet Source                                                                                                                                  | <1 % |
| 29 | <a href="http://www.academia.edu">www.academia.edu</a><br>Internet Source                                                                                                                                      | <1 % |

# Table of Contents

| Chapter                                                                | page |
|------------------------------------------------------------------------|------|
| Chapter 1 .....                                                        | 1    |
| 1.1 Project Domain:.....                                               | 2    |
| 1.2 Problem Identification:.....                                       | 2    |
| 1.2.1 Proposed Solution:.....                                          | 2    |
| 1.2.2 Objectives:.....                                                 | 2    |
| 1.2.3 Scope of the Project:.....                                       | 3    |
| 1.3 Effectiveness / Usefulness of the System: .....                    | 3    |
| 1.4 Resource Requirement: .....                                        | 3    |
| 1.4.1 Hardware: .....                                                  | 3    |
| 1.4.2 Software: .....                                                  | 3    |
| 1.4.3 Data: .....                                                      | 3    |
| 1.5 Report Organization: .....                                         | 4    |
| Chapter 2 .....                                                        | 6    |
| 2.1. Introduction .....                                                | 7    |
| 2.2 Explanation of Important Constructs of the Application Domain..... | 7    |
| 2.2.1 Post-Stroke Speech Disorders .....                               | 7    |
| 2.2.2 Speech Therapy and Rehabilitation.....                           | 8    |
| 2.2.3 AI and NLP in Speech Therapy.....                                | 8    |
| 2.2.4 Mobile Health (mHealth) Solutions and Gamification.....          | 8    |
| 2.3. Existing studies/systems.....                                     | 9    |
| 2.3.1 Related Systems/Applications .....                               | 11   |
| 2.4.1. Identified Problems from Existing Work .....                    | 14   |
| 2.5. Summary .....                                                     | 14   |
| Chapter 3 .....                                                        | 15   |
| 3.1 Introduction .....                                                 | 16   |
| 3.2 Interface Requirements.....                                        | 16   |
| 3.2.1 Hardware Interface Requirements .....                            | 16   |
| 3.2.2 Software Interface Requirements .....                            | 16   |
| 3.3 Functional Requirements.....                                       | 17   |
| 3.4 Use Case Model.....                                                | 17   |
| 3.5 Use Cases .....                                                    | 19   |
| 3.5.1 Use Case 1: User Registration and Login.....                     | 19   |
| 3.5.2 Use Case 2: Therapy Session.....                                 | 19   |
| 3.5.3 Use Case 3: Progress Tracking.....                               | 19   |
| 3.6 Non-Functional Requirements.....                                   | 20   |
| 3.7 Resource Requirements.....                                         | 20   |

|                                                                   |    |
|-------------------------------------------------------------------|----|
| 3.8 Database Requirements .....                                   | 20 |
| 3.9 Project Feasibility .....                                     | 20 |
| 3.9.1 Technical Feasibility .....                                 | 21 |
| 3.9.2 Operational Feasibility .....                               | 21 |
| 3.9.3 Legal and Ethical Feasibility .....                         | 21 |
| 3.10 Summary .....                                                | 21 |
| Chapter 4 .....                                                   | 22 |
| 4.1 Introduction .....                                            | 23 |
| 4.2 System Design .....                                           | 23 |
| 4.3 Design Approach .....                                         | 23 |
| 4.4 Interface Design .....                                        | 24 |
| 4.5 4+1 View Model of Architecture .....                          | 25 |
| 4.5.1 Logical View (Class Diagram) .....                          | 25 |
| 4.5.2 Process View (Activity, State, and Sequence Diagrams) ..... | 27 |
| 4.5.3 Development View (Component Diagram) .....                  | 32 |
| 4.5.4 Physical View (Deployment Diagram) .....                    | 33 |
| 4.7 Summary .....                                                 | 34 |
| Chapter 5 .....                                                   | 35 |
| 5.1 Introduction .....                                            | 36 |
| 5.2 Modules of the Project .....                                  | 36 |
| 5.2.1 Audio Preprocessing Module .....                            | 36 |
| 5.2.2 Deep Learning Model Module .....                            | 36 |
| 5.2.3 Flask API Module .....                                      | 37 |
| 5.2.4 Flutter Mobile Application Module .....                     | 37 |
| 5.3 System Workflow Summary .....                                 | 38 |
| 5.4 Summary .....                                                 | 39 |
| Chapter 6 .....                                                   | 40 |
| 6.1 Introduction .....                                            | 41 |
| 6.2 Testing Setup and Environment .....                           | 41 |
| 6.2.1 Hardware and Software Environment .....                     | 41 |
| 6.2.2 Testing Methodology .....                                   | 42 |
| 6.3 Functional Testing .....                                      | 43 |
| 6.3.1 Authentication Module .....                                 | 43 |
| 6.3.2 Patient Dashboard Module .....                              | 45 |
| 6.3.3 Exercise Module .....                                       | 47 |
| 6.3.4 Therapist Dashboard Module .....                            | 49 |
| 6.4 Speech Recognition Accuracy Testing .....                     | 51 |
| 6.4.1 GRU-Based Speech Clarity Prediction .....                   | 51 |
| 6.4.2 Speech Recognition Test Results .....                       | 52 |

|                                                           |    |
|-----------------------------------------------------------|----|
| 6.4.3 Analysis of Speech Recognition Results .....        | 54 |
| 6.5 Firebase Integration Testing .....                    | 54 |
| 6.5.1 Firestore Database Operations.....                  | 54 |
| 6.6 Performance Testing.....                              | 57 |
| 6.6.1 App Performance Metrics.....                        | 57 |
| 6.6.2 Battery Consumption.....                            | 57 |
| 6.7 Usability Testing .....                               | 58 |
| 6.7.1 Test Setup .....                                    | 58 |
| 6.7.2 Usability Feedback Analysis .....                   | 58 |
| 6.8 Cross-Platform Compatibility .....                    | 59 |
| 6.10 Summary .....                                        | 60 |
| Chapter 7 .....                                           | 62 |
| 7.1 Introduction .....                                    | 63 |
| 7.2 Review Against Proposed Objectives.....               | 63 |
| 7.2.1 Objectives Comparison .....                         | 63 |
| 7.3 Key Achievements .....                                | 64 |
| 7.3.1 Real-Time Speech Grading System.....                | 64 |
| 7.3.2 Dual-Role Architecture .....                        | 65 |
| 7.3.3 Session Management System .....                     | 65 |
| 7.3.4 Exercise Variety .....                              | 65 |
| 7.3.5 Progress Tracking and Reporting .....               | 65 |
| 7.3.6 Firebase Backend Integration .....                  | 65 |
| 7.4 Limitations.....                                      | 66 |
| 7.4.1 Dependency on Internet Connection .....             | 66 |
| 7.4.2 Speech Recognition in Noisy Environments .....      | 66 |
| 7.4.3 Single Language Support .....                       | 66 |
| 7.4.4 No Audio Playback of Patient’s Spoken Phrases ..... | 66 |
| 7.4.5 No Video or Visual Feedback .....                   | 67 |
| 7.5 Future Recommendations.....                           | 67 |
| 7.5.1 Offline Mode with Local Storage.....                | 67 |
| 7.5.2 Audio Playback for Patient’s Spoken Phrases.....    | 67 |
| 7.5.3 Multi-Language Support .....                        | 67 |
| 7.5.4 Visual Articulation Guides .....                    | 68 |
| 7.5.5 Session Analytics Dashboard for Therapists.....     | 68 |
| 7.5.6 iOS Deployment.....                                 | 68 |
| 7.5.7 Accessibility Features.....                         | 68 |
| 7.6 Summary .....                                         | 69 |
| REFERENCES .....                                          | 71 |

## List of Figures

|                                            |    |
|--------------------------------------------|----|
| Figure 3.1 Use Case Diagram.....           | 18 |
| Figure 3.2 Patient Use Case.....           | 19 |
| Figure 3.3 Therapist Use Case.....         | 19 |
| Figure 4.1 Class Diagram.....              | 26 |
| Figure 4.2 Patient Activity Diagram .....  | 28 |
| Figure 4.3 Therapist Activity Diagram..... | 29 |
| Figure 4.4 State Diagram.....              | 30 |
| Figure 4.5.1 Sequence Diagram .....        | 31 |
| Figure 4.5.2 Sequence Diagram .....        | 32 |
| Figure 4.6 Component Diagram.....          | 33 |
| Figure 4.7 Deployment Diagram.....         | 34 |

## List of Tables

|                                                   |    |
|---------------------------------------------------|----|
| Table 2.1 Comparison of existing systems.....     | 12 |
| Table 6.1 Hardware and Software Environment ..... | 41 |
| Table 6.2 Authentication Module.....              | 43 |
| Table 6.3 Patient Dashboard Module .....          | 45 |
| Table 6.4 Exercise Module.....                    | 47 |
| Table 6.5 Therapist Dashboard Module .....        | 49 |
| Table 6.6 Speech Recognition Results .....        | 53 |
| Table 6.7 Firestore Database Operations.....      | 54 |
| Table 6.8 App Performance Metrics .....           | 57 |
| Table 6.9 Cross Platform Compatibility.....       | 59 |
| Table 7.1 Objectives Comparison .....             | 63 |

# **Chapter 1**

# **INTRODUCTION**

Speech problems after a stroke are really tough for people. They make it hard to talk to others. This can make people feel lonely, frustrated and sad. Many people, with speech problems after a stroke feel embarrassed or hesitant to get help. They worry about what others might think. They find it hard to get to therapy sessions.

## **1.1 Project Domain:**

This project is about using Artificial Intelligence in Healthcare and making a Mobile Application. This app uses tools like speech recognition and Natural Language Processing to give people feedback on how they are saying words. It also makes sure that people can use it easily on their phones.

## **1.2 Problem Identification:**

People recovering from a stroke and working on speech problems have some big problems. The patients feel uncomfortable seeing a speech therapist in public and so they avoid therapy. Speech therapy may also cost a lot of money. Many patients have to see a therapist quite often, and this can be difficult for them. The other big issue is the typical approach to therapy that fails to provide feedback to people, and they have to wait to see how they are doing. Over time the exercises can get boring. Individuals become unmotivated to continue practicing.

### **1.2.1 Proposed Solution:**

We want to solve these issues by developing a phone application to improve the speech of patients with stroke. The app analyzes people's speech patterns, and assists in saying words correctly by using Artificial Intelligence. It also has games and awards to help make practicing fun and motivating. The app provides feedback allowing individuals to see how they're doing. It is a suitable alternative to therapy and the individuals can practice at home without feeling embarrassed.

### **1.2.2 Objectives:**

The main objectives of this project are:

- To make an app that uses Artificial Intelligence to give people feedback on their speech.
- To add games and Rewards to keep people motivated.
- To make sure the app is easy to use for adults by designing a simple UI.

### **1.2.3 Scope of the Project:**

That project is focused on creating exercises that will help individuals with their speech, in terms of saying words and speaking smoothly. The app offers tools to assess individuals' speaking skills and to improve them. The app is being developed to run on Android devices, which will be used by people. Games are also added to make it more playful.

## **1.3 Effectiveness / Usefulness of the System:**

The software being developed has a number of advantages when compared to speech therapy. Individuals can practice at home at any time. They receive immediate feedback to help them get better more quickly. People want to continue training since the games make it more enjoyable. Moreover, as fewer visits have to the therapist, it also saves money.

## **1.4 Resource Requirement:**

### **1.4.1 Hardware:**

To use the app, users must have Android phones or tablets. To manage all the data and ensure the app functions properly, a computer server will be utilized.

### **1.4.2 Software:**

We are using Python, Flutter, and other tools and technologies to develop the app. We also make use of the TensorFlow and Natural Language Processing to make artificial intelligence work. All the data is processed with Firebase, and we need a Google speech recognition tool.

### **1.4.3 Data:**

The app requires a significant amount of data to operate properly. We have some speaking exercises for them to use and we gather data from their use of the app

to provide them with feedback. We also monitor user progress, allowing us to improve the app.

## **1.5 Report Organization:**

There are six chapters in this report. In **Chapter 1**, the problem is identified, an introduction is provided and a suggested solution is presented. The research is summarized and existing speech treatment solutions and AI-based approaches are explained in **Chapter 2**. **Chapter 3** deals with methodology, which includes system development, implementation of AI, mobile application design, etc. The implementation of the system is discussed in **Chapter 4**, where the focus is on the design of the user interface, development of the back end, and integration of the AI model. **Chapter 5** deals with testing and assessment, as well as a discussion of how the system performs, is used and applies to the real world. The report is provided in **Chapter 6** which contains tests testing the limitations, analysis and results of the project. **Chapter 7** ends with a summary of the project's achievements, future recommendations and targets.



**Chapter 2**  
**BACKGROUND AND EXISTING**  
**SYSTEMS**

## **2.1. Introduction**

This chapter offers a more detailed explanation of the current research and systems in mobile health, artificial intelligence used in rehabilitation and the speech therapy for post-stroke patients. It familiarises the reader with key concepts needed to grasp the project and reviews previous work and existing technologies. This chapter aims to establish conceptual knowledge and awareness of what has already been done with speech rehabilitation technologies and the shortcomings that justify a need for an AI-powered mobile health solution. Knowledge of prior research allows us to define how the proposed system will add to the accessibility, accuracy and engagement levels for adults after stroke

## **2.2 Explanation of Important Constructs of the Application Domain**

Developing an effective AI-based speech therapy solution requires a deep understanding of speech disorders, rehabilitation methods, artificial intelligence techniques, and mobile health technologies. This section explains these concepts in detail so that the overall domain becomes clear and meaningful.

### **2.2.1 Post-Stroke Speech Disorders**

A stroke that damages the part of the brain that controls speech can result in the person experiencing speech difficulties. Aphasia, apraxia of speech, and dysarthria are the most common. Aphasia results in difficulties in understanding and speaking, leading to problems in word and sentence formulation. Apraxia of speech is caused by the brain's inability to control mouth and tongue movement although the muscles are not weak. Dysarthria, however, is caused by weak muscles used for speaking, making the patient's speech slow, unclear or slurred. These speech disorders make speaking more difficult. Feeling frustrated or avoiding social situations may delay recovery. Recognising these disorders is crucial so that treatment is directed to solve their particular communication problems.

### **2.2.2 Speech Therapy and Rehabilitation**

Good speech and language is important after stroke and speech therapy plays a crucial role in recovering. Face-to-face therapy with a speech language pathologist is called traditional therapy. In these sessions they might read or repeat words, move their mouths, or complete language tasks. This therapy is effective but traditional therapy requires people to travel to clinics to see their therapists which for many adults is difficult due to distance, money and reluctance. So, telehealth (aka telemedicine) and home therapy are on the rise. Using these technologies allows patients to practise at their own pace and makes therapy more continuous. This is crucial to improve the recovery after stroke as recovery must be ongoing.

### **2.2.3 AI and NLP in Speech Therapy**

New applications of artificial intelligence (AI) and Natural Language Processing (NLP) in speech therapy have emerged. AI speech recognition systems can transcribe speech and detect pronunciation, fluency and articulation errors. NLP can also process speech to discern improvements or deterioration in speech. This technology is used to build systems that can provide therapeutic feedback. A number of studies demonstrate the beneficial effects of AI tools on motor and cognitive function in stroke. They allow for objective assessment, eliminate the need for constant therapist intervention, and enable therapy at any time. But there are some challenges surrounding the limited databases of disordered speech and privacy concerns.

### **2.2.4 Mobile Health (mHealth) Solutions and Gamification**

Mobile health (mHealth) apps are a common approach to provide healthcare on mobile platforms. Speech therapy mobile apps can be used to do exercises while on the move, interactive speech lesson and progress monitoring. Studies have shown that mobile rehabilitation can help patients improve their language skills and gain more self-confidence. Gamification is also useful, that is, the use of game-like elements such as rewards, points, levels and challenges. Gamification helps with adherence and the gamified aspects make it fun.” It is particularly relevant for people with stroke as it may impact outcomes.

## 2.3. Existing studies/systems

In the field of speech language pathology, artificial intelligence and digital therapy platforms have made great strides. Researchers are interested in the rehabilitation of speech impairment. There are different approaches to the development of such systems for post-stroke patients: AI-based speech recognition, adaptive therapy systems, mobile applications.

In a study in 2024 the author in [1] reviewed 704 studies on the use of AI in adult stroke rehabilitation, and categorized the studies into four main areas: treating impairments, assistive interventions, prediction/imaging, and neuroscience. They found that AI tools such as robotics, brain–computer interfaces, wearable sensors and machine learning techniques can help improve motor, walking and cognitive recovery, especially for upper-limb rehabilitation, and could make therapy more personalized. However there are challenges such as the lack of standard datasets, high device costs, privacy concerns and the “black-box” nature of AI models hinder its widespread adoption.

The author in [2] analyzed 29 studies on mobile applications for stroke rehabilitation and classified them into 5 groups, namely therapy applications, educational applications, rehab videos, reminder applications, and hybrid types. Therapy apps helped with motor weakness and language problems, and reminder and video apps helped patients stick with exercises, they found. Many studies, however, had a small number of participants, a short follow-up time (approximately 1 month) and a weak study design, making it more difficult to draw strong conclusions.

In 2024 the author in [3] reviewed the process of rehabilitating stroke. They wrote about diagnostic techniques, acute and rehabilitation interventions (such as physical, occupational, speech and cognitive therapy) and new technologies (such as neurostimulation, robotics and virtual reality). They said their review was a summary, not new information, but they stressed the future is about early treatment, individualised treatment and treatment that utilises brain plasticity.

VerbalizVR in [4] is a mobile VR application for post-stroke aphasia that includes modules of speech, listening, cognitive exercises, and real-world practice. In a small pilot (n=17) of patients and therapists, users reported high satisfaction and no safety issues were identified. However, the study was short term with a small sample and the app was only a speaking skills app with limited real life scenarios so it is not yet proven to be effective.

The author in [5] conducted a review of 169 studies involving 6,404 stroke patients on gamified rehabilitation devices, such as robotic/motorized tools, non-motorized devices, virtual reality systems, and electrical stimulation. These gamified tools generally enhanced motor and cognitive recovery and increased emotional, social and motivational outcomes with high satisfaction in most patients. Most studies were small or early stage, however, larger and more standardized trials are needed.

A study in [6] was performed on the use of AI in serious games and gamification for health, including 16 studies, mostly related to rehabilitation for motor or cognitive problems. Machine learning was the most common AI approach, and direct AI integration into games was rare, they found. Early results showed high usability and motivation and patient satisfaction. However, the studies were mostly small pilot studies with few participants. More convincing evidence is needed to make any firm conclusions.

A review 35 studies of tele-speech-language therapy for post-stroke aphasia in [7] found 40% of the interventions to be live video, 57% to be self-administered programs and 3% to be both. They are more about training or therapy than assessment. However, 77% of the studies were of low quality and there is a lack of research in areas such as remote assessment and live therapy, pointing to the need for better designed trials and guidelines.

A web and smartphone app in [8] called VerbalizAPP®, for early post-stroke aphasia therapy. In a pilot study of 40 patients, starting around 5 days post-stroke, language scores were significantly improved after three months with very positive feedback from patients and no side effects. However, this was a small single center study without a control group and larger controlled trials are needed to confirm the effectiveness.

A proposed multicenter randomized trial of 76 subjects in [9] comparing a personalized digital speech therapy app to a workbook for post-stroke dysarthria was described in which the patients will practice 1 hour/day for 4 weeks. The primary outcome is speech clarity. The study will also examine feasibility and adherence, but these outcomes are not yet available. The app is currently only in Korean and does not blind participants.

A review on therapeutic robots for post-stroke rehabilitation in [10] found that robotic devices can greatly help motor and cognitive recovery, reduce therapists' workload, and improve patient outcomes through precise and repeatable training. The authors suggested combining these robots with virtual reality or telemedicine and emphasized that standardizing their use could improve long-term recovery results.

This study in [11] examined guideline adherence in speech and language therapy following stroke. Ninety percent of patients reported communication and/or swallowing difficulties but only 44% received the recommended outpatient therapy . Older age and longer hospital stay were associated with higher likelihood of receiving guideline-based care . Female sex was associated with lower likelihood of diagnosis and longer therapy . The authors identified care gaps and suggested improved monitoring to tackle inequities.

### **2.3.1 Related Systems/Applications**

A range of mobile and PC applications are available to help people with speech disabilities. The most prominent are:

**Tactus Therapy Suite:** It is an integrated app with multiple modules for language therapy, articulation therapy and cognitive therapy. It has a lot of features but could be overwhelming for a new user, especially without the support of a therapist.

**Constant Therapy:** It provides computer-adapted therapy based on performance. It includes telemonitoring by therapists but no speech correction.

**Speech Assistant AAC:** An augmentative and alternative communication (AAC) device for non-verbal people to help them siturate. But it lacks exercises and progress monitoring.

Our proposed system differs by offering real-time AI-powered feedback and engaging gamified modules for adult users recovering from stroke.

## Table 2.4 Comparison of existing systems:

*Table 1 : 2.1 Comparison of existing systems*

| Sr No. | Paper Name /Applications                                                                                                               | Authors / Platform         | Strengths, Features & Purpose                                                                                                                                                                                                                                        | Limitations                                                                                                                                            |
|--------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | AI Applications in Adult Stroke Recovery and Rehabilitation: A Scoping Review                                                          | Isuru Senadheera et al.    | Reviewed 704 studies to see how AI is used in stroke recovery. Found 4 main areas: physical recovery, assistive devices, prediction/imaging, and brain research. Shows AI can help in therapy, especially for arm recovery, robotics, and brain-computer interfaces. | No standard datasets, high cost of AI tools, and privacy concerns. Models are often hard to explain.                                                   |
| 2      | Effect of Mobile Application Types on Stroke Rehabilitation: A Systematic Review                                                       | Stephen G. Szeto et al.    | Looked at 29 studies on mobile apps for stroke recovery. Found 5 types of apps: therapy, education, videos, reminders, and mixed types. Therapy apps improved movement and speech. Reminder/video apps helped patients stick to exercises.                           | Many studies had small groups and weak research design. Short follow-up (about 1 month). Hard to compare because apps and results were very different. |
| 3      | Stroke Rehabilitation: From Diagnosis to Therapy                                                                                       | Xiaohong Li et al.         | Gives an overview of stroke treatment from start to finish. Covers physical, speech, and cognitive therapies, plus new tech like brain stimulation and assistive devices. Stresses early treatment and future personalized rehab.                                    | Not a detailed research review—based on selected papers, so it might miss some studies. No numbers or data analysis included.                          |
| 4      | A Mobile-based Virtual Reality Speech Rehabilitation App for Patients With Aphasia After Stroke: Development and Pilot Usability Study | Xiaofan Bu et al.          | Created a VR app for speech therapy in people with aphasia. App has 4 modules: speaking, understanding, thinking, and real-life tasks. Tested with 17 people (patients, healthy people, staff, and VR experts) and got good feedback.                                | Only tested on a small group, no proof it works long-term. Focuses on speaking, not reading/writing. VR content is limited.                            |
| 5      | Gamified Devices for Stroke Rehabilitation: A                                                                                          | Juan J. Sánchez-Gil et al. | Reviewed 169 studies with over 6,000 patients using game-based rehab devices                                                                                                                                                                                         | Many studies were small or not high quality. Devices,                                                                                                  |

|    |                                                                                   |                          |                                                                                                                                                                                                                                                                                 |                                                                                                                                                                       |
|----|-----------------------------------------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Systematic Review                                                                 |                          | (robots, VR, non-motorized tools, electrical stimulation). Found they improve movement, thinking skills, and motivation. Patients enjoyed them and felt more engaged.                                                                                                           | methods, and results varied too much to compare well. Few studies measured “fun” factor in a standard way.                                                            |
| 6  | The Role of AI in Serious Games and Gamification for Health                       | Daniel Tolks et al.      | Looked at studies using AI with health-related games. Most focused on movement recovery (like after stroke) and some on thinking skills. Shows AI + games can keep patients motivated and improve therapy.                                                                      | Most studies were small, early tests. Not many directly used both AI and games together. Field is still new, and better-quality studies are needed.                   |
| 7  | Current Status of Tele-speech Language Therapy (Scoping Review)                   | Yuhei Kodani et al.      | Scoping review of 35 studies on tele-speech therapy for post-stroke aphasia. Classified as synchronous (e.g., videoconference), asynchronous (e.g., automated training), or combined. Mostly asynchronous. Calls for more research on remote assessment and real-time training. | Quality of included studies mostly low. Did not use standard quality checklists. Only included English/Japanese papers—possible language/publication bias.            |
| 8  | Digital Tool as Speech and Language Therapy for Patients with Post-stroke Aphasia | Gerardo Ruiz Ares et al. | Developed VerbalizAPP®, a smartphone/web app for early post-stroke aphasia. In a pilot of 40 patients, language scores improved significantly. High patient satisfaction and usability. Can be used in hospital and at home.                                                    | Pilot before-after study with no control group (due to COVID). Small sample (40) and single site. Only short-term results measured. Larger multicenter trials needed. |
| 9  | Efficacy and Feasibility of a Digital Speech Therapy... (RCT Protocol)            | Yuyoung Kim et al.       | Multicenter RCT protocol for post-stroke dysarthria. Compares personalized speech app vs workbook therapy (60 min/day, 5 days/week, 4 weeks). 76 patients planned, acute to chronic phases. Main outcome: speech intelligibility; also measures speech and well-being.          | Protocol only—no results yet. Small planned sample (76). App in Korean only. No participant blinding. Possible dropout in older patients due to tech issues.          |
| 10 | Therapeutic Robots for Post-stroke Rehabilitation (Review)                        | Ronghua Hong et al.      | Review of robotic therapy devices. Summarizes how robots can automate training, reduce therapist workload, and improve motor/cognitive recovery. Highlights robots’ advantages in training intensity and feedback compared to standard PT/OT.                                   | Narrative review—not a systematic meta-analysis. Relies on varied-quality published studies. No new experimental data. Possible publication bias.                     |
| 11 | Guideline Adherence in Speech and Language                                        | Daniel Schindel et al.   | Claims-data study of 4,486 German stroke patients. Found only 44.1% received outpatient SLT. Women and older patients less likely to get full                                                                                                                                   | Uses insurance billing data, which may under-report. No data on private insurance holders. Content and                                                                |

|    |                             |             |                                                                                |                                                                                |
|----|-----------------------------|-------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
|    | Therapy in Stroke Aftercare |             | recommended care. Highlights gaps in aftercare and need for better monitoring. | quality of therapy unknown.                                                    |
| 12 | Tactus Therapy Suite        | Android/iOS | Modules for articulation, language, and cognitive training                     | Complex interface, better suited for clinical use, High one-time upfront cost. |
| 13 | Constant Therapy            | Android/iOS | AI-adaptive exercises, therapist monitoring                                    | Monthly subscription cost, Limited real-time correction                        |
| 14 | Speech Assistant AAC        | Android/iOS | AAC tool for non-verbal users                                                  | No rehabilitation or therapy tracking                                          |

### 2.4.1. Identified Problems from Existing Work

Based on an analysis of the existing speech therapy tools and systems, there are some main limitations:

**Therapist dependency:** Professional supervision is often needed for customization and feedback in applications, causing therapist dependency.

**Limited engagement:** No gamification, low motivation and adherence.

**Poor accessibility:** There are some platforms that may cost a lot and some platforms will need complex procedures to access.

**Non-specific targeting:** Most of these apps have been created for kids or for a particular disorder - but not for adult post-stroke rehab.

The difficulties emphasize the significance of creating a mobile answer that incorporates availability, immediate criticism, and captivating interaction.

## 2.5. Summary

In this chapter, we provided a thorough definition of basic terms related to speech therapy, aphasia, artificial intelligence, and mHealth. It discussed key publications and systems, demonstrating the use of technology for speech therapy. The table offers a better insight of the benefits and limitations of systems available. From the review, it is clear many of the existing systems do not provide real-time feedback based on AI, or personalisation, or provide an interesting therapy approach. This has led to the development of a better solution.

## **Chapter 3**

# **Requirements Specification**

## **3.1 Introduction**

This chapter describes in detail the requirements of the proposed AI-powered mobile application for post-stroke speech therapy. It explains the hardware, software, functional, and non-functional needs of the system to ensure that the development and deployment of the application are effective. The chapter also includes the use case model, use case descriptions, resource requirements, database needs, and project feasibility study. The aim is to develop a clear understanding of how different parts of the system interact with each other and how they fulfill the goal of helping stroke patients in speech rehabilitation through a simple and accessible mobile solution.

## **3.2 Interface Requirements**

Interface requirements specify how different parts of the system will interact with each other and with external systems. As the proposed project involves both hardware and software parts, this section is divided into two parts: hardware interface requirements and software interface requirements.

### **3.2.1 Hardware Interface Requirements**

The system will be implemented on smart devices such as mobile phones or tablets having android 10.0 or above, 2GB of RAM and 32GB of storage. The mobile device has an inbuilt microphone to capture the user's voice. There will be speakers or headphones to give the audio cues. Internet (Wi-Fi or mobile data) access would be required to transmit speech data for processing and provide immediate feedback.

### **3.2.2 Software Interface Requirements**

The system will be built using an Android based application called Flutter. The cloud backend will be entirely handled through Firebase for authentication, real-time data handling, and structured data storage. Speech recognition will be done using Google Speech-to-Text API, AI processing will be done using TensorFlow and speech analysis will be done with the help of NLP libraries. The user interface is simplified and easy to use for the patients and caregivers.

### **3.3 Functional Requirements**

The proposed mobile application will help in some of the key functions. It will provide real-time speech analysis for pronunciation, intonation, and fluency. There will be therapy exercises in the app that will make it interactive. All users will have a secure and personalized profile. Data will be stored securely and retrieved when necessary.

### **3.4 Use Case Model**

The use case model depicts the interaction of the various types of users with the system. The key actors of this system are post-stroke patients, caregivers and therapists. Patients will perform speech therapy exercises via the app, caregivers will be able to assist patients with the app, and therapists will be able to monitor progress remotely.

The use case diagram of this project shown in Figure 3.1 will be used to show the basic functionalities like login for the user, therapy sessions, feedback generation, and progress tracking.

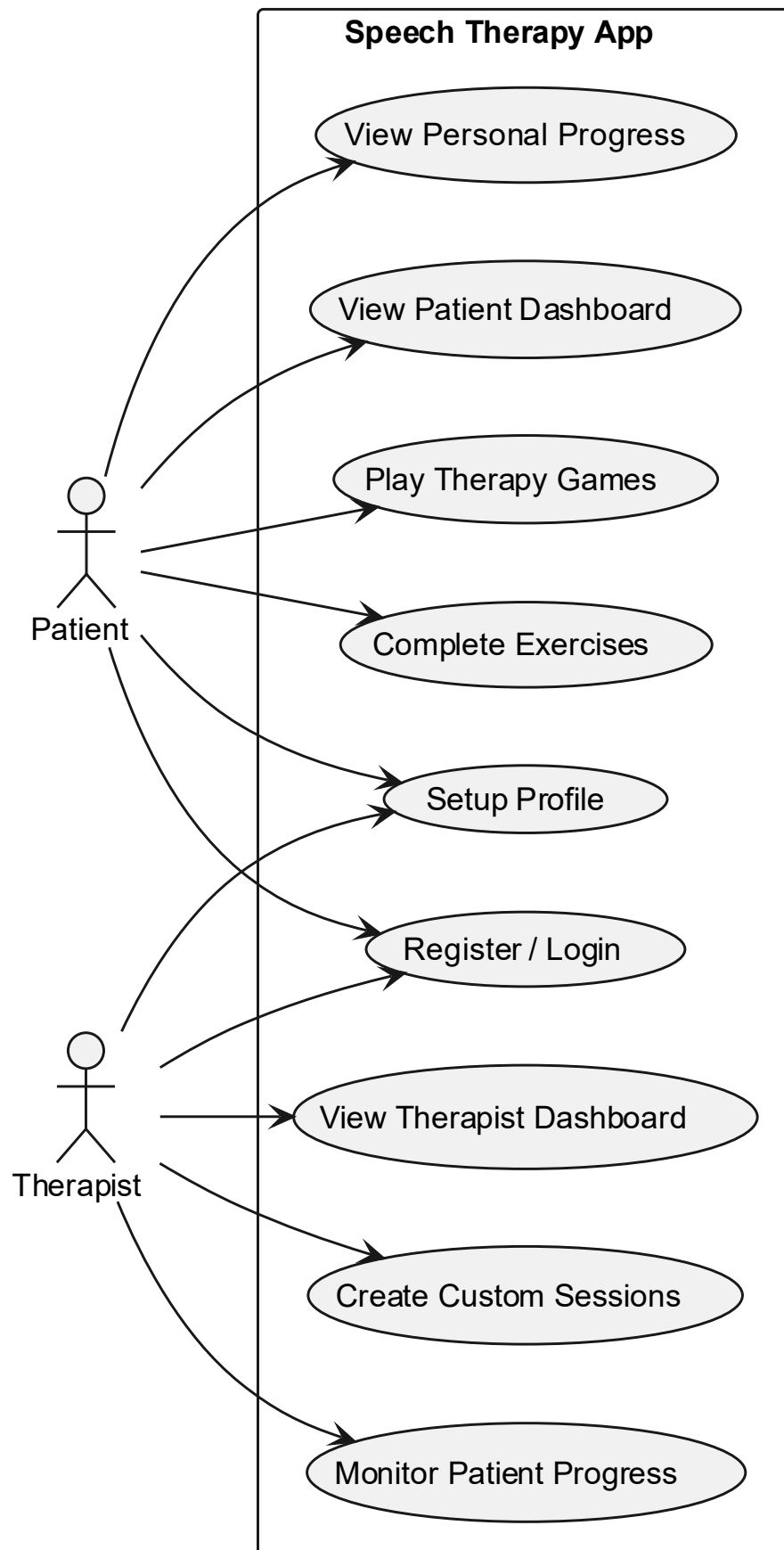


Figure 1 : 3.1 Use Case Diagram

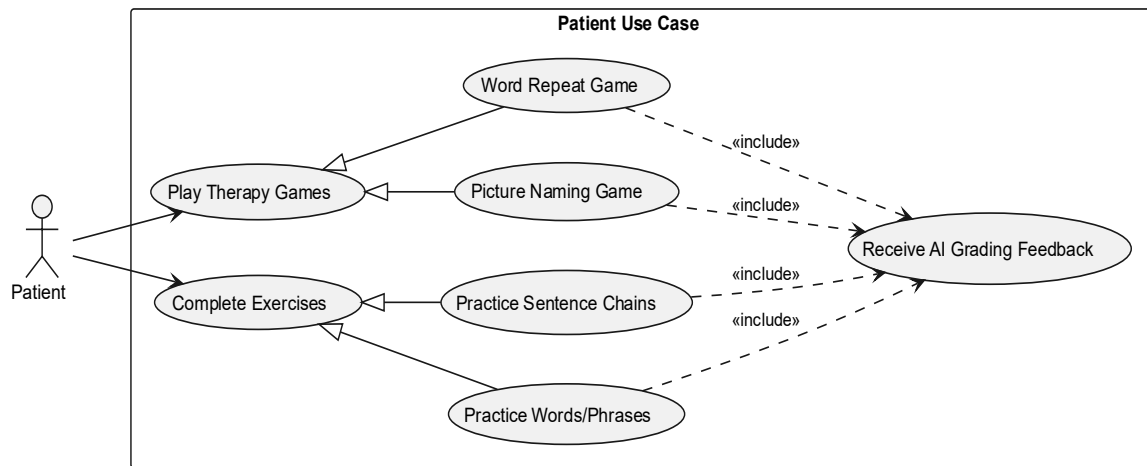


Figure 2 : 3.2 Patient Use Case

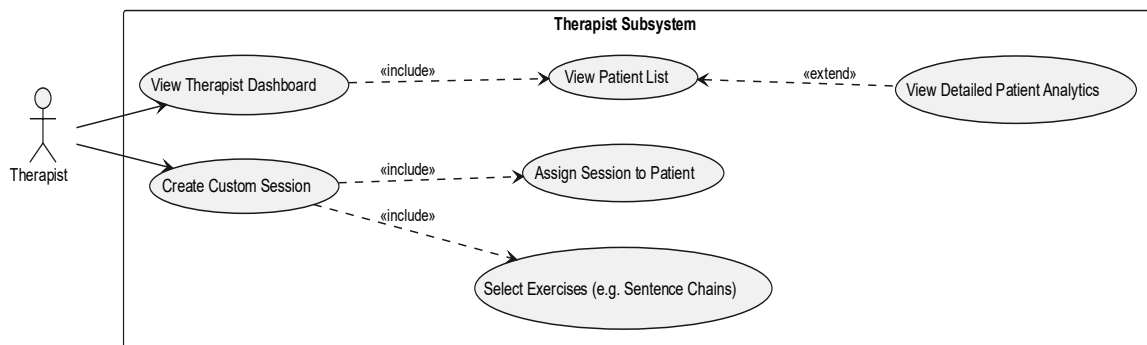


Figure 3 : 3.3 Therapist Use Case

## 3.5 Use Cases

This section describes the main use cases of the system in detail.

### 3.5.1 Use Case 1: User Registration and Login

First, the users of the system, patients, caregivers, and therapists, will register and then log into the application using the firebase authentication. Before logging in they will need to download the app on a mobile device Then they can log in and have access to therapy sessions and other tailored features.

### 3.5.2 Use Case 2: Therapy Session

The patient will select a therapy module, perform speech exercises with real-time feedback. This will guide the patient to improve their speech slowly.

### 3.5.3 Use Case 3: Progress Tracking

The application will show the patient's performance data and therapy history. This will assist patients, caregiver and therapists to track improvement over time and plan future therapy sessions accordingly.

### **3.6 Non-Functional Requirements**

Non-functional requirements define the quality standards of the system.

- Provide speech analysis and feedback within three seconds to keep the process smooth and interactive.
- Support up to 500 users simultaneously without performance issues.
- Work reliably across all supported devices without crashing.
- Ensure all data is encrypted during storage and transmission for security.
- Provide a uniform user experience across various Android devices and screen sizes.

### **3.7 Resource Requirements**

The project is dependent on the availability of smartphones to test, development laptops equipped with Flutter, Python programming language, and TensorFlow, and cloud accounts for Firebase. The human resources required are an AI developer, a mobile app developer, a UI/UX designer and a project manager to oversee and coordinate the whole process.

### **3.8 Database Requirements**

User authentication, real time data, and structured user data like speech session data and performance tracking data will be managed in Firebase.

### **3.9 Project Feasibility**

Feasibility analysis to see if the project is feasible (technically, operationally and ethically). It guarantees the proposed solution can be developed within the available resources, works efficiently in real world conditions and satisfies all professional standards. This analysis is useful to confirm also the sustainability, cost-effectiveness and long-term viability of this system.

### **3.9.1 Technical Feasibility**

The project is rooted in widely available technologies which can be operated using common Mobile phones. It delivers the real-time performance through APIs and cloud services, without the need for expensive hardware. Technically viable, simple to deploy, and being implemented with lightweight mobile processing and Firebase-based backend integration, it's appropriate for consumers with fundamental gadgets.

### **3.9.2 Operational Feasibility**

The app is created in an easy and accessible way, so that elderly users with limited technical skills will be able to use the app. Going to a therapy session, getting feedback and seeing progress is an easy and effective process. It is easy to use, clear and user friendly. The users will not be confused to use the system and also will not request the external guidance repeatedly.

### **3.9.3 Legal and Ethical Feasibility**

The system will adhere to all privacy laws and will not retain personal data without the user's consent. It is only developed for therapeutic purposes and it will not harm its users. All data will be securely held and only accessible to authorized persons. Full confidentiality will be maintained. The app also abides by ethical guidelines to protect user trust and avoid any misuse of sensitive information.

## **3.10 Summary**

This chapter presented a complete specification of the requirements for the proposed speech therapy application. It identified the hardware and software requirements, functional and non-functional requirements and the feasibility aspects of the project. These specifications provide a solid basis for the development of a secure, accessible and efficient mobile application to support the rehabilitation of stroke patients, providing reliable performance and long-term usability.

## **Chapter 4**

# **SYSTEMS MODELLING**

## **4.1 Introduction**

This chapter discusses the design and architecture of our proposed project, “Post-Stroke Speech Therapy for Adults.” The system modelling is used to describe structure, behavior and flow of data in the application. It helps developers, designers and stakeholders to understand how different parts of the system will interact. The diagrams and models in this chapter illustrate how users, processes and data interact in carrying out speech therapy tasks. This chapter is intended to give a clear picture of the system structure before actual coding and implementation.

## **4.2 System Design**

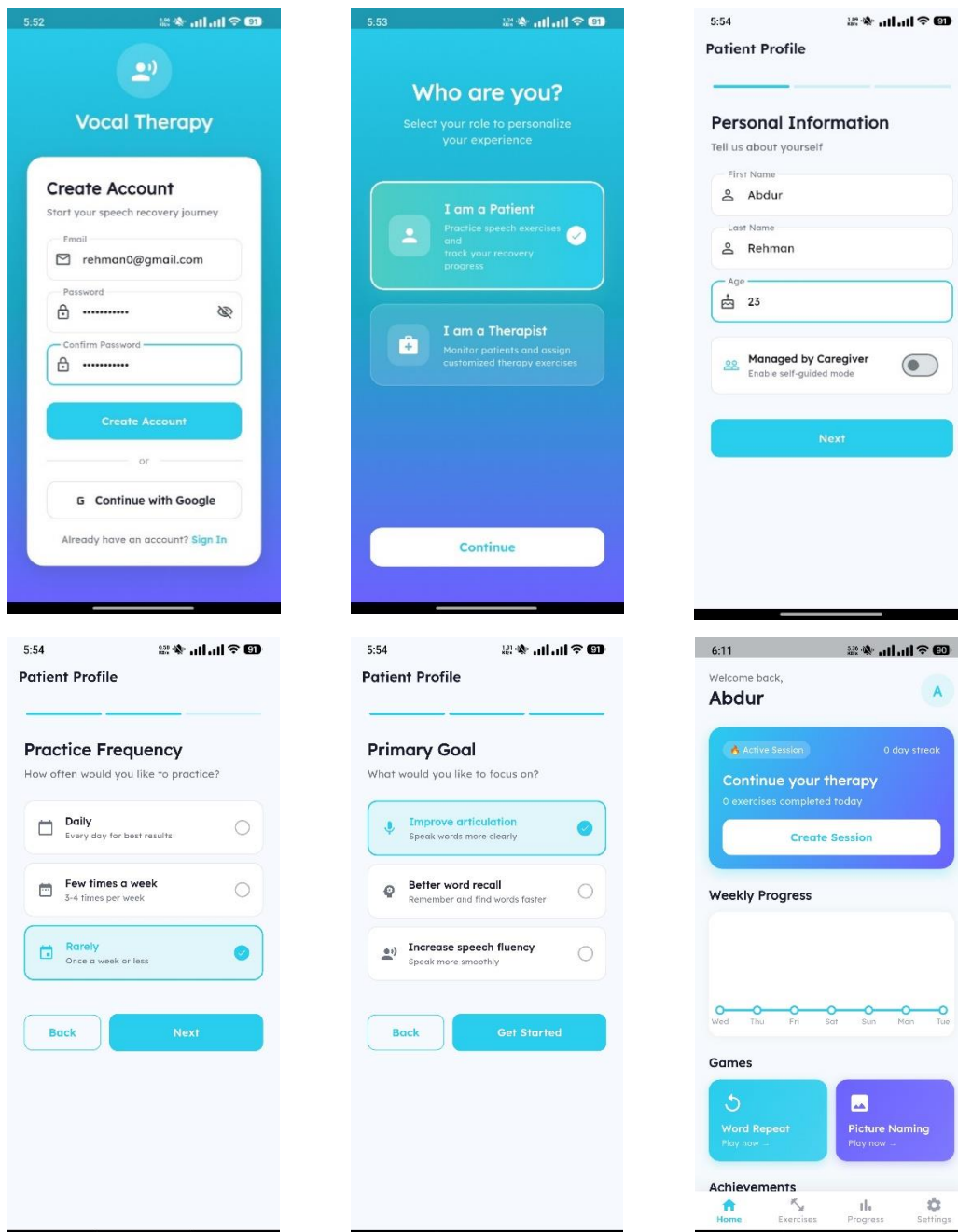
It is a system designed to create an easy to use mobile app that will enable stroke patients to do speech therapy exercises at home. The system consists of speech recognition, generation of feedback, and keeping track of progress. The backend utilizes AI and natural language processing to analyze speech and provide feedback and the frontend offers a simple and easy to use interface for users. All data is stored securely in a cloud database and users/caregivers can view progress anytime. The primary objective of the design is to develop an efficient, secure and user friendly system for all ages.

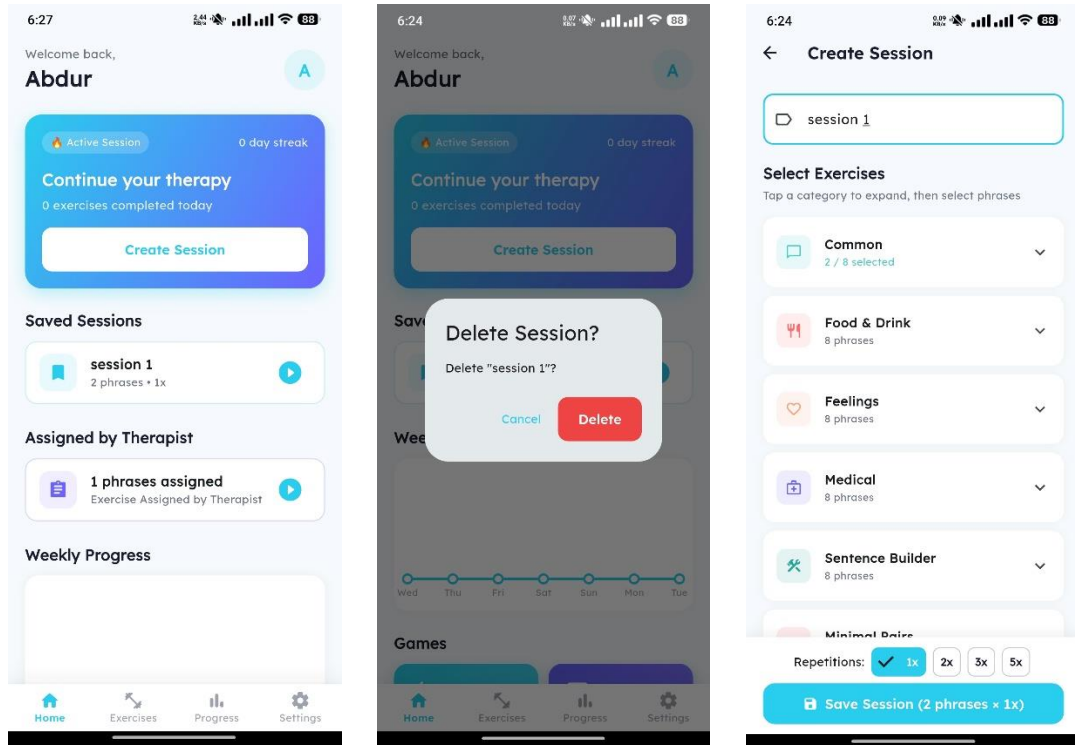
## **4.3 Design Approach**

There are two different design patterns that are used in Software Development: Top-Down Design and Bottom-Up Design. Our system uses the Bottom-Up Design approach. The method used in this approach is to break the system down to smaller and independent parts and then assemble the whole system. For the speech analysis portion of our project, we began with the selection of datasets, pre-processing, and the training of AI models. Once that was done, we built the mobile application UI, integrated with Firebase for database management, and added individual features like therapy sessions, individual speech recognition, feedback generation and tracking progress. Lastly, the integration of all these modules was carried out step-by-step to create the whole Post Stroke Speech Therapy system. This has also been applied to each individual component and to ensure that the components of the system perform well together.

## 4.4 Interface Design

The mobile application interface is designed to be easy and convenient for older users who might not have technology literacy as high as a younger person. On the home screen you will see options such as therapy sessions, progress reports and feedback history. The big buttons and icons are easy to read. The soothing professional colour scheme allows the user to relax. Audio guided exercises, on screen instructions to help you along the way. The overall design of the app allows the user to move around the app easily.





## 4.5 4+1 View Model of Architecture

The system architecture is described by the 4+1 View Model which describes the system from different views namely logical, process, development and physical views. This model helps to understand the overall system and the interaction between each component.

### 4.5.1 Logical View (Class Diagram)

The logical view is the structure of the system in terms of classes and their relationships. The class diagram is made up of classes like User, Session, Exercise, Feedback, Progress and so forth. Each class has different methods and attributes. For instance, the User class contains information on login and profile information. The Session class keeps track of the exercises that the user performs. The connections between these classes are the paths that data takes through the system. One User can have multiple sessions, and each session can include multiple exercises, which record feedback and progress information.

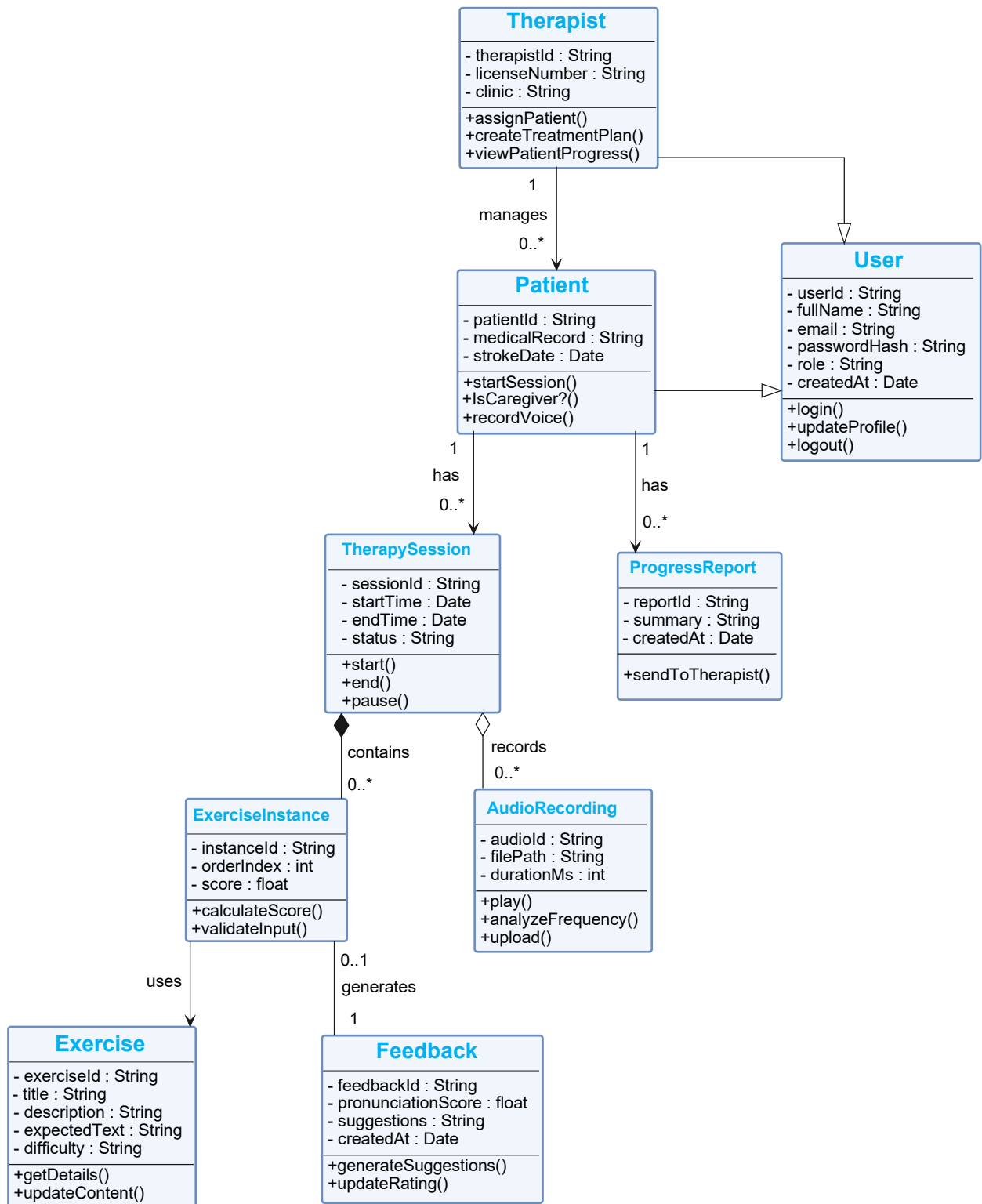


Figure 4 : 4.1 Class Diagram

## **4.5.2 Process View (Activity, State, and Sequence Diagrams)**

Process view describes the behavior of the system in response to a user's interaction. It explains the interactions and manipulation of data by various parts of the system in real time. The activity diagram describes how the actions including login, start of therapy session, speech recording, input processing, and the output of feedback are executed. The state diagram describes the way the app transitions from one state to another, such as from idle to recording, recording to processing, and processing to feedback display. The sequence diagram illustrates the interaction and flow of messages between the user, mobile interface, and AI backend. Combined these diagrams show the overall working and coordination of the system.

### **4.5.2.1 Activity Diagram:**

This is a flow diagram of the user journey for the speech therapy app. It starts with the entry into the App, selection of the therapy module and selection of exercises. Then, the user goes through the process of recording their voice, having it analyzed by AI and getting feedback. The user can repeat exercises or finish the session, after which the results are saved, and achievements are awarded. This diagram helps visualize the entire therapy session process from start to finish.

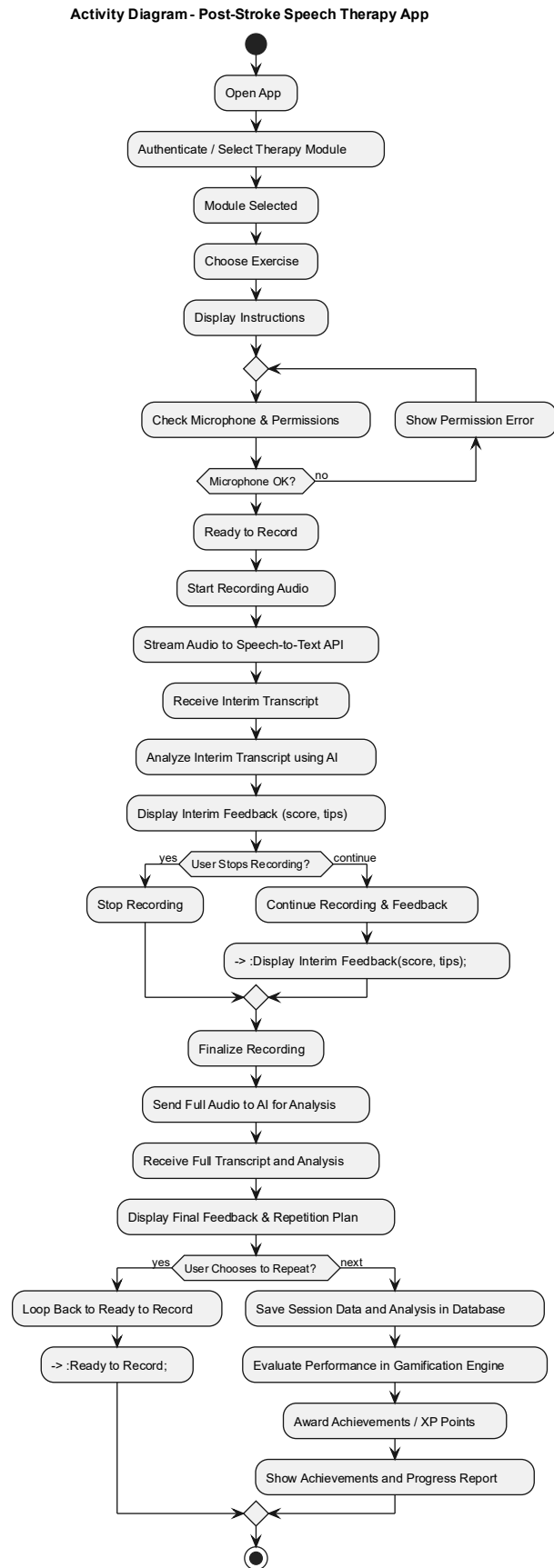


Figure 5 : 4.2 Patient Activity Diagram

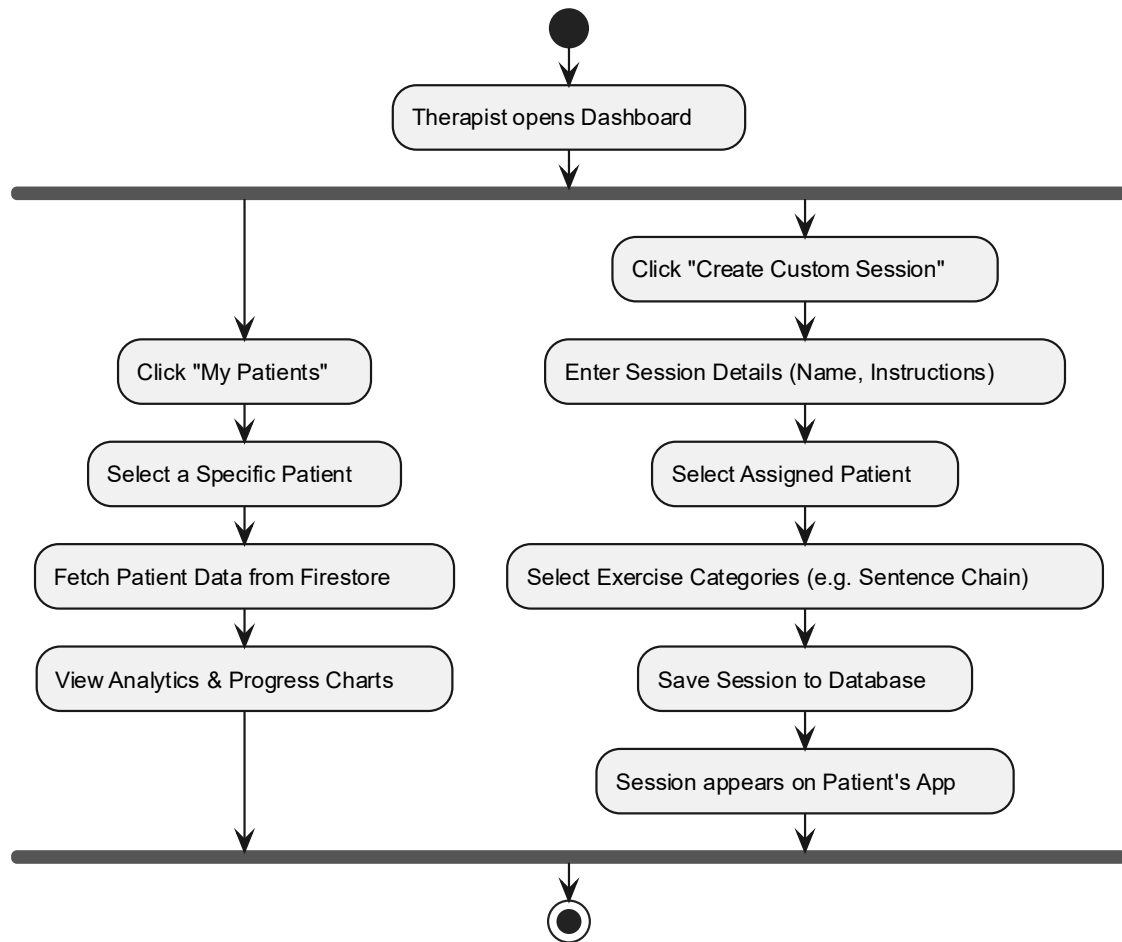
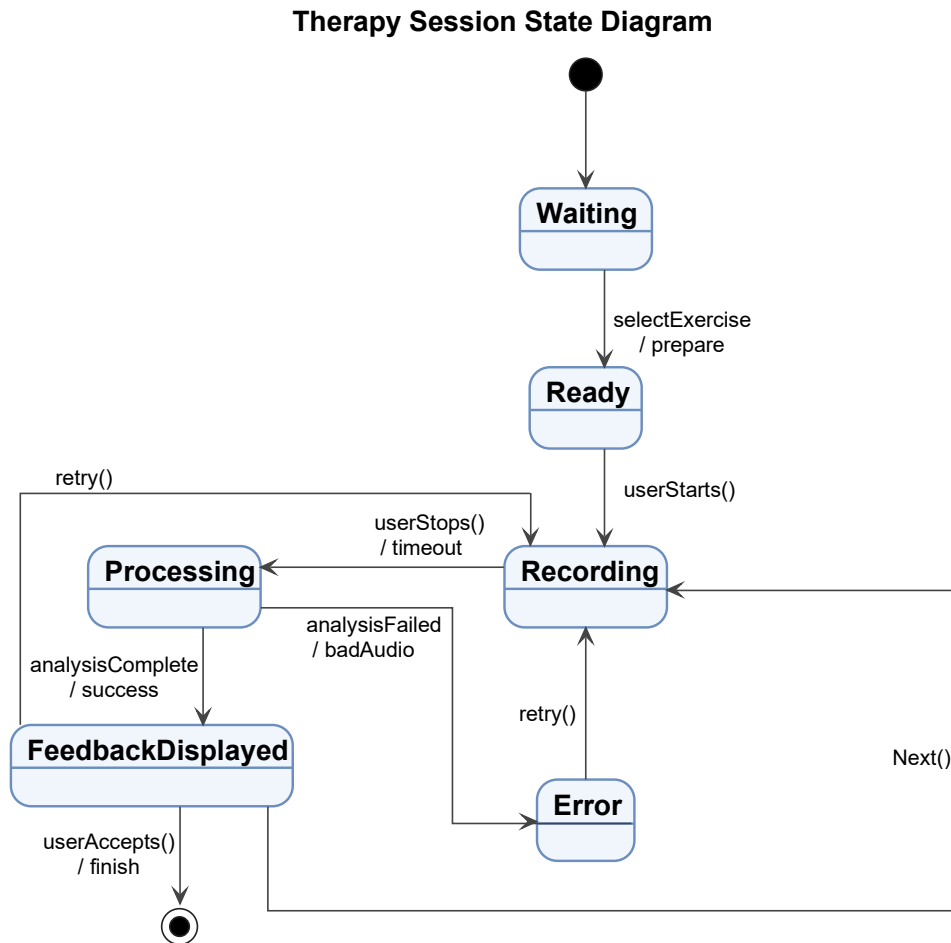


Figure 6 : 4.3 Therapist Activity Diagram

#### 4.5.2.2 State Diagram:

The state diagram shows the different states of a therapy session and the transition of the session among these states. First it is in a waiting state, then it goes through ready, recording and processing states. The session then proceeds to feedback, paused, completed or error states depending on the result. Finally the session can be persisted. The diagram shows clearly the lifecycle of a therapy session and the possible transitions.



*Figure 7 : 4.4 State Diagram*

#### 4.5.2.3 Sequence Diagram

##### 1. User Registration & Login Sequence:

This sequence diagram shows the steps taken when a user registers and logs into the application. It involves communication between user, mobile app, authentication service (Firebase), database and email service. The process involves creating user account, verifying email and getting user profile at the time of login. The above diagram clearly shows how the system controls user authentication and registration.

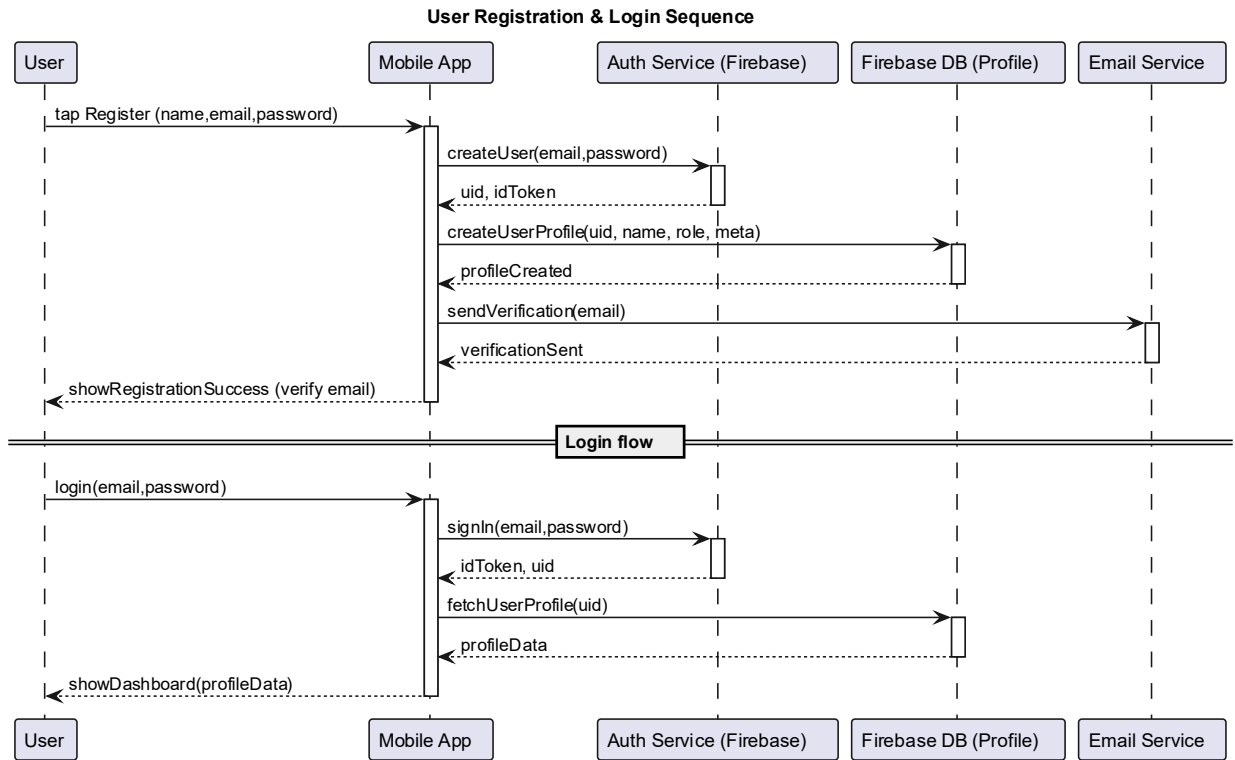


Figure 8 : 4.5.1 Sequence Diagram

## 2. Real-Time Therapy Session Sequence:

This diagram shows how the patient, mobile app, audio recorder, speech-to-text API, AI engine, database, and gamification engine work together in real-time during a therapy session. It shows the process of audio recording and streaming, transcription, analysis and giving feedback in a continuous way. Finally the session data is stored and achievements are calculated. It helps to understand the dynamic communication flow among various components.

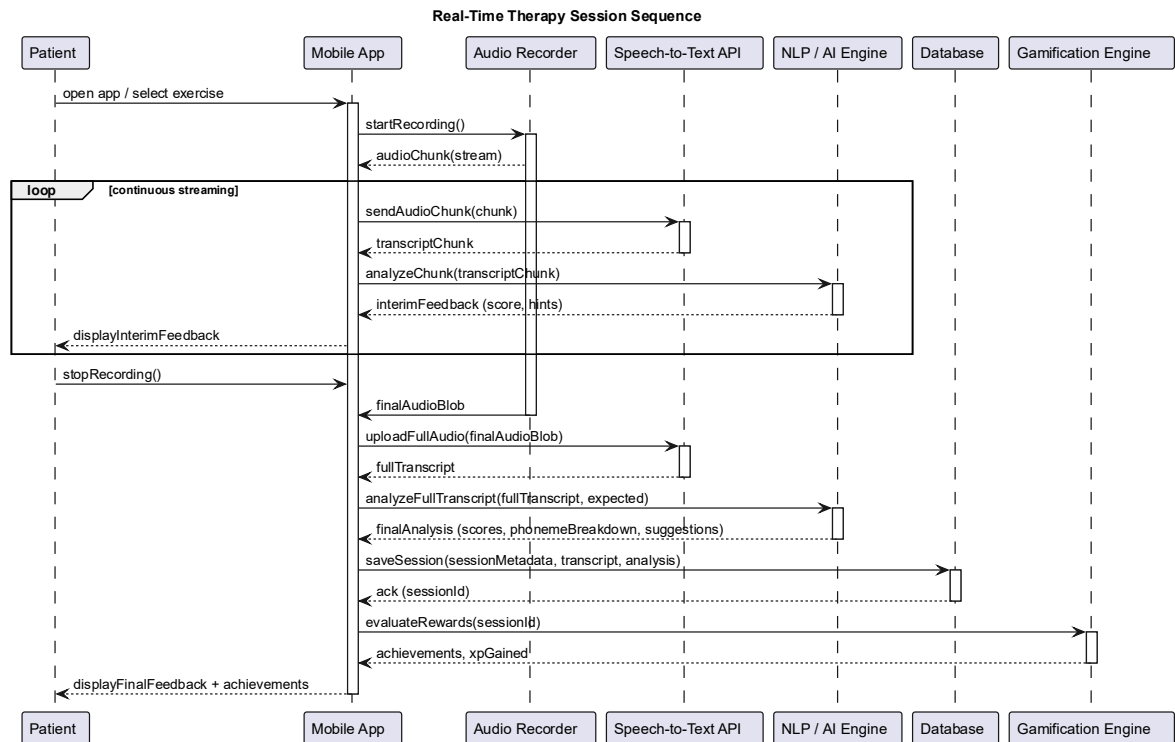
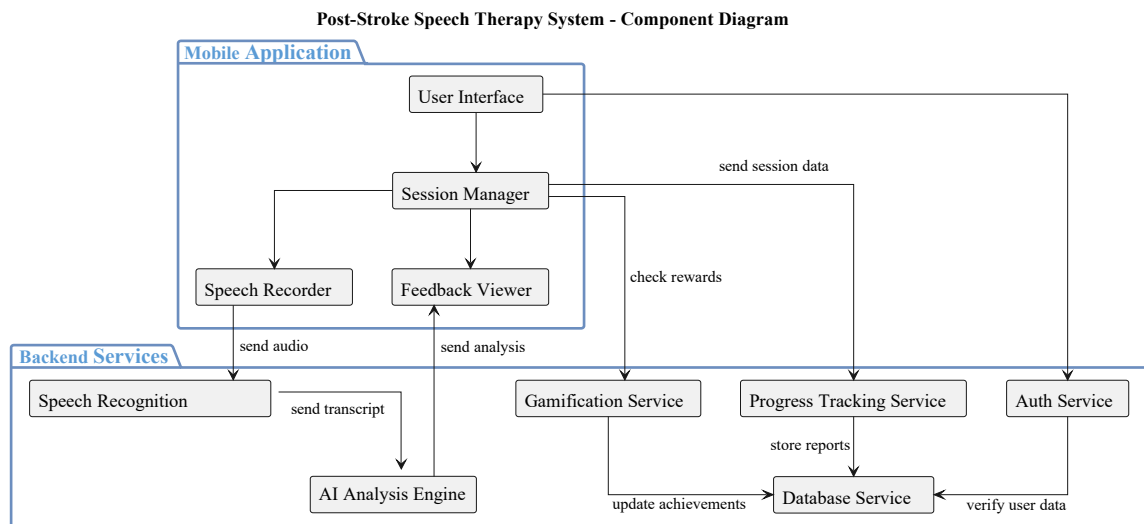


Figure 9 : 4.5.2 Sequence Diagram

### 4.5.3 Development View (Component Diagram)

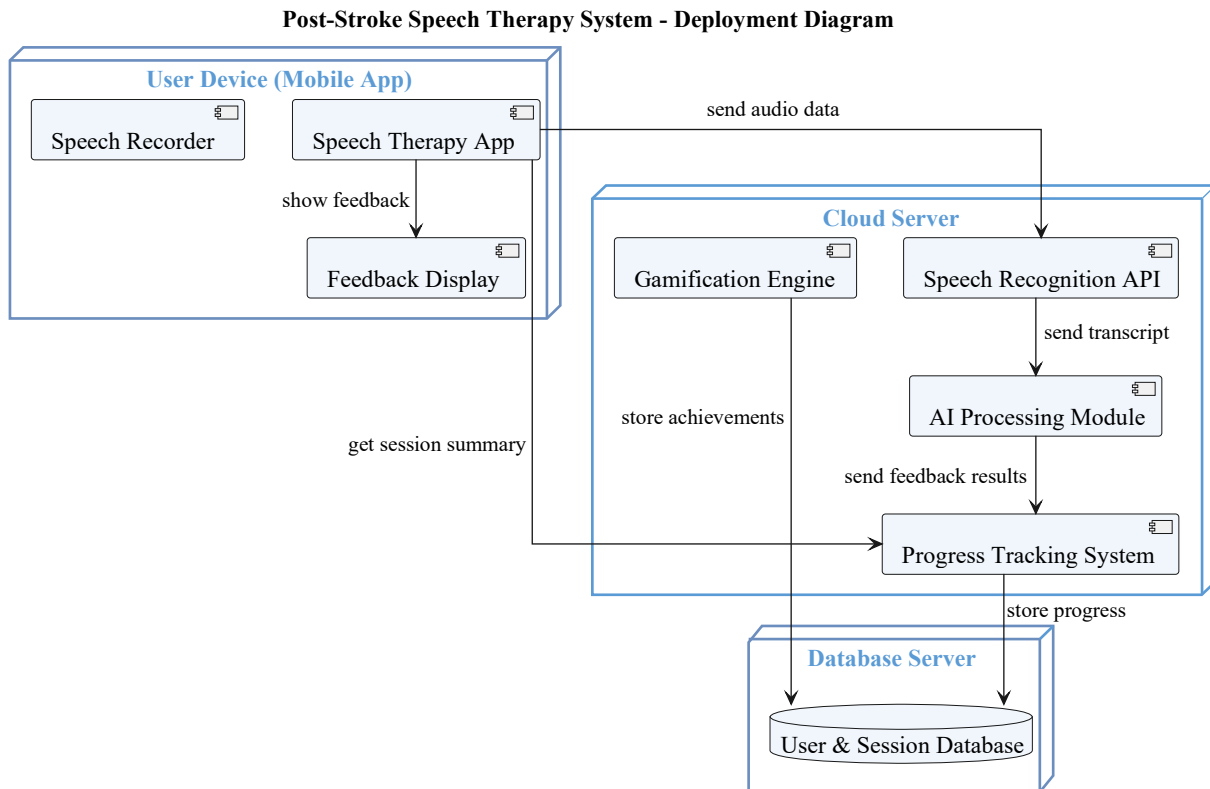
The development view shows how the system is partitioned into different software components. The main elements are mobile application frontend, AI processing module, speech recognition system and database. The mobile app is developed using Flutter and the AI module is developed using Python and Tensorflow, NLP libraries. The speech recognition is done by Google Speech-to-Text API. The database stores all user and session information. All these parts work together to keep the system running well and efficiently.



*Figure 10 : 4.6 Component Diagram*

#### 4.5.4 Physical View (Deployment Diagram)

The physical view describes the hardware and network architecture used to deploy the system. The system consists of user mobile devices with the installed app, a cloud server for the AI model and backend services, and a database server for managing user data. These elements are interconnected via the Internet . They can communicate and synchronize in real time . This gives users the ability to access the system whenever they want and get immediate feedback on their speech performance.



*Figure 11 : 4.7 Deployment Diagram*

## 4.7 Summary

In this chapter, the design and modelling of the system have been discussed in detail. It explains how different components of the “Post-Stroke Speech Therapy for Adults” application work together. The system design, interface design, and architectural views show the logical and functional structure of the project. The 4+1 view model presents the system from multiple angles, while the ERD helps in understanding the database relationships. This chapter provides a complete understanding of the system’s internal design, which will help in the development and implementation of the project.

## **Chapter 5**

# **IMPLEMENTATION**

## 5.1 Introduction

This chapter explains how the proposed project 'Post Stroke Speech Therapy for Adults' will be implemented. Designed for patients after stroke to improve the clarity of their speech with the help of AI and mobile exercises for therapy. This translates to a smart and intuitive platform for therapy, supported by machine learning algorithms, speech understanding, API development and mobile app development.

The system has four components: audio preprocessing, deep learning GRU model, a flask API to serve the model, and a flutter mobile application for user interactions. The structure, behavior, and data flow of the application was represented using system modelling techniques. The diagrams and modules in this chapter describe the communication between users, processes and data to carry out speech therapy functions. This chapter aims to offer a clear understanding of the design and implementation of the system prior to its deployment.

## 5.2 Modules of the Project

### 5.2.1 Audio Preprocessing Module

This module extracts numerical features from raw audio recordings that can be interpreted by the deep learning model. The module was written in Python 3.13.3, with the following libraries: Librosa, NumPy, and SoundFile. The audio from users can have different lengths, quality or sampling rate and thus needs to be preprocessed prior to training and prediction. The audio files are converted to monaural and resampled to 16 kHz to keep consistency.

Signals of speech are preprocessed and Mel Frequency Cepstral Coefficients (MFCCs) are extracted from the signals as they represent important features of speech. The audio recordings are transformed into a sequence of 13 MFCC features. Each recording of sound is converted to a sequence of 13 MFCC features. The audio recordings are of varying lengths and the MFCC sequences are consecutively padded or truncated to ensure that they have a fixed sequence length before being fed to the GRU model. The extracted features are then normalized for better model stability and prediction. All speech recordings are

fed in a consistent format into this pre-processing stage for a sequential deep learning analysis.

### **5.2.2 Deep Learning Model Module**

The central part of the system is a deep learning model implemented in TensorFlow and Keras which is currently based on GRU (Gated Recurrent Unit). Speech clarity is predicted by the model on a scale ranging from 0 (not at all clear) to 100 (very clear). Over 1000 speech audio samples were taken and scored for speech clarity. With the dataset provided, both clear speech as well as slurred speech recordings from the post-stroke patients were included. GRU networks differ from typical Artificial Neural Networks (ANNs) as they are able to capture sequential and temporal patterns from the data, which makes them more appropriate for speech analysis since speech signals carry information over time.

The model designed and implemented includes an input layer to handle a sequence of MFCC features, GRU layers to process the sequences of speech, dense layer with ReLU activation functions, and the output layer with a linear activation function to predict the clarity score. Adam optimization, Huber Loss function and Mean Absolute Error (MAE) were used for training and evaluation of the model. The hyperparameters of the training process were: 50 epochs, batch size: 32 and the train-validation split was 80/20. The GRU architecture enhanced the speech clarity prediction performance of the system by improving its understanding of patterns, fluency, and speech flow of the speech.

### **5.2.3 Flask API Module**

To enable communication between the deep learning model and the Flutter mobile app, a REST API was created with Flask. Flask was used to create a REST API for the communication between deep learning model and Flutter mobile application. The Flask API is the intermediary that links the mobile app and the AI model. There are two API endpoints available: “/” is for testing the server connection and “/analyze” is for speech analysis and prediction.

The API preprocesses the input audio by resampling and computing MFCC features before a user uploads an audio recording using the app. The processed features are then fed to the GRU model to predict the speech clarity score. The API will then use Google Speech-to-Text services to convert the spoken audio into text, after making the prediction. The recognized speech is compared to the expected speech to provide feedback for the user.

Finally, the API returns a response in JSON that includes the score for the forecast clarity, recognized text, expected phrase, feedback message and model confidence. The module allows to easily communicate between the mobile application and AI system, and the speed and accuracy of speech analysis.

#### **5.2.4 Flutter Mobile Application Module**

Front end is built in mobile application using flutter and dart language, a simple and easy to use interface to the patient affected by stroke. Flutter was selected because it can facilitate cross-platform mobile application development, and provide seamless integration with backend APIs. The user will go through the following screens: login, register, user dashboard, selection of exercises to listen to, the recording of the user's voice, and a results screen.

Secure login functionality was implemented using Firebase Authentication and user progress and therapy history were stored using Cloud Firestore. Inside the app, users may record their voices and send them to the Flask API via an HTTP POST request. The application outputs the speech clarity score, recognized speech and feedback in simple, understandable format after processing. The interface is designed to be easily accessible and simple for the patients with stroke. The Flutter app also works very efficiently on Android devices, bringing about speech therapy to more people.

### **5.3 System Workflow Summary**

The system workflow starts with the user logging in to the Flutter mobile app and choosing a speech exercise. The user speaks into the user mobile app and the recorded speech is uploaded to the Flask API.

The API processes the audio data to obtain MFCC features and resampling to preprocess it. The features are then fed into the deep learning model, which is based on GRU, that predicts speech clarity score. The API also leverages the Google Speech-to-Text services to convert speech to text and to compare the recognized speech to the expected phrase. Based on the comparison, feedback is generated for the user.

Lastly, the mobile app shows the clarity score, identified text, therapy feedback and records the user's improvement on the database. This will allow users to obtain feedback on their speech production in real time and ongoing supervision for therapy to gradually improve their speech intelligibility.

## **5.4 Summary**

The implementation of the system of “Post-Stroke Speech Therapy for Adults” proposed in the previous chapter was explained. The project integrates audio preprocessing, deep learning, backend API development and mobile application technologies to provide a fully integrated speech therapy solution.

The audio preprocessing module preprocessed speech signals and extracted MFCC features, which are more suitable for deep learning analysis. A new model using a GRU-based deep learning architecture was used instead of the former ANN model to model the sequential characteristics of speech and enhance the accuracy of speech clarity prediction. The use of a Flask API ensured that there was proper communication between the AI model and the Flutter mobile application, and the mobile application was user-friendly. Overall, the system was designed to provide an intelligent, efficient, and user-friendly speech therapy platform capable of assisting post-stroke patients in improving their speech abilities through AI-powered feedback and analysis.

In summary, the system aimed to create a smart, efficient, and user-friendly speech therapy platform that leverages AI to help post-stroke patients enhance their speech skills.

## **Chapter 6**

# **Results, Testing, Analysis and Validation**

## 6.1 Introduction

This chapter presents the testing, analysis, and validation of the Post-Stroke Speech Therapy system. The purpose of this chapter is to verify that the proposed system meets its intended objectives and performs correctly under different conditions. Each module of the system was tested individually as well as part of the complete integrated application. The testing process includes the evaluation of the deep learning model, accuracy of the speech clarity prediction, functionality of the Flask API, the integration with Firebase, the responsiveness of the UI, and usability of the mobile application.

The Post-Stroke Speech Therapy system has been created, with the mobile app written in Flutter, the back-end API services in Flask, the GRU-based deep learning model in TensorFlow and Keras, and authentication and data storage in Firebase. On speech analysis, audio preprocessing techniques are used with MFCC feature extraction. Speech is then fed into the GRU model to produce a score for speech clarity, and Google Speech-to-Text services for speech-to-texting and feedback generation. The application is used by patients after stroke to practice their speech and get automated feedback about the clearness and pronunciation of the patient's speech.

Android testing was done in Android devices with Android studio emulator, along with physical mobile devices. In this chapter all test results are based on real runs of applications, audio recordings and evaluation data collected during the testing period.

### 6.2.1 Hardware and Software Environment

The following hardware and software were used during the testing phase:

*Table 2 : 6.1 Hardware and Software Environment*

| Component                | Details                                   |
|--------------------------|-------------------------------------------|
| Development Machine      | Windows 10/11 PC, Intel Core i5, 8 GB RAM |
| Mobile Device (Physical) | Android smartphone (Android 12+)          |

|                    |                                                  |
|--------------------|--------------------------------------------------|
| Emulator           | Android Studio Emulator (Pixel 6, API 33)        |
| Framework          | Flutter 3.9.2 (Dart language)                    |
| Backend            | Firebase (Authentication, Cloud Firestore)       |
| Speech Recognition | Google Speech-to-Text via speech_to_text package |
| Grading Algorithm  | Longest Common Subsequence (LCS)                 |
| IDE                | Android Studio / VS Code                         |
| Version Control    | Git                                              |

### 6.2.2 Testing Methodology

The system was tested through various testing methods to guarantee reliability, accuracy and usability.

**Unit Testing:** Individual modules and functions were tested separately like MFCC feature extraction, audio preprocessing functions, GRU model prediction functions, Flask API endpoints, and Firebase database operations.

**Integration Testing:** Different modules were tested for their interactions, including Mobile App → Flask API → Audio Preprocessing → GRU Model Prediction → Feedback Generation → Firebase Storage.

**Functional Testing:** All the features and screens of the mobile app were tested manually to ensure the system performs as per the requirements.

**Usability Testing:** The application was tested with a small group of users who gave feedback on the interface and usability of the application, as well as user experience..

**Performance Testing:** API response time, model prediction time, app startup time, screen transition time and Firebase synchronization performance.

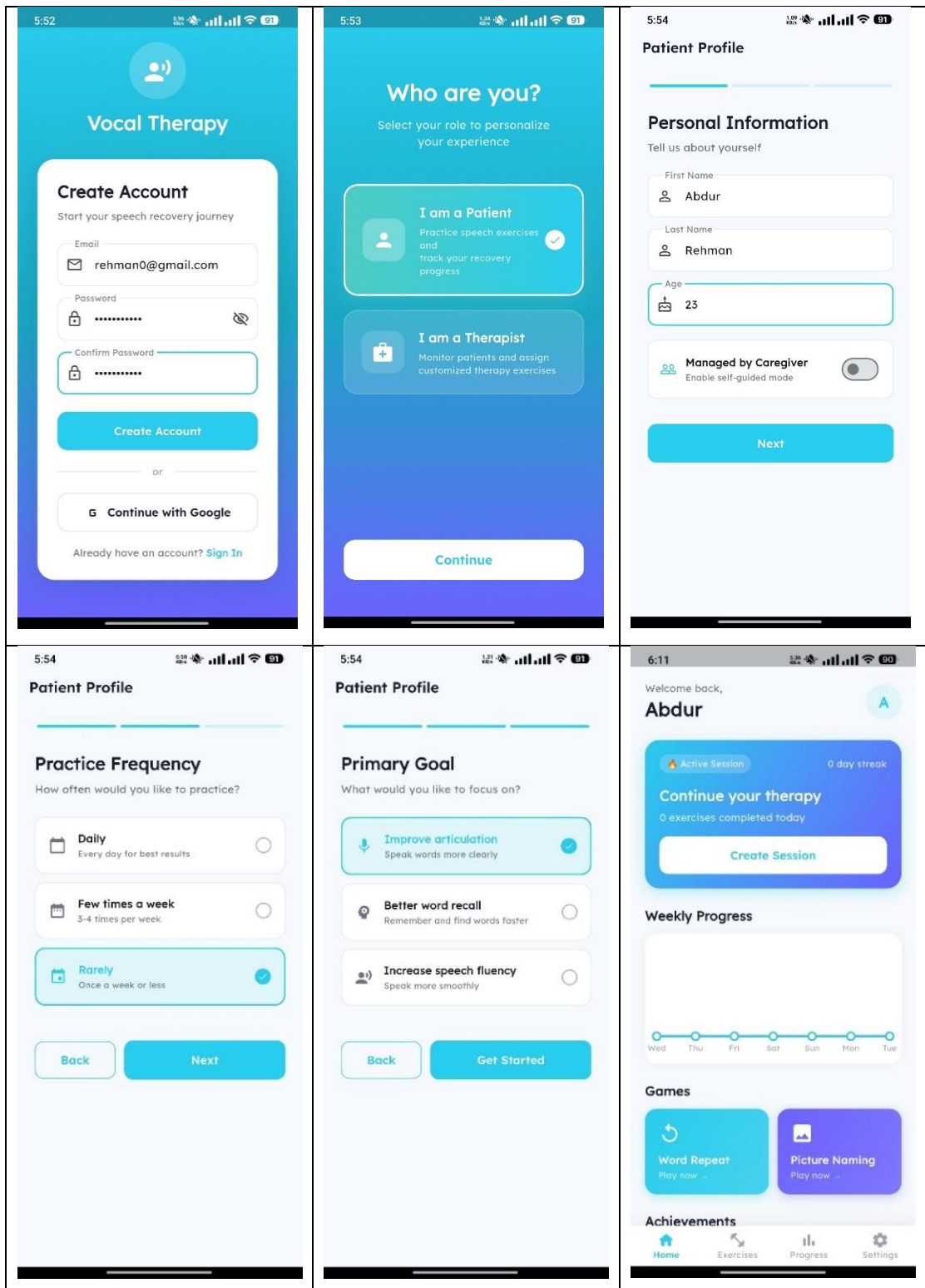
## 6.3 Functional Testing

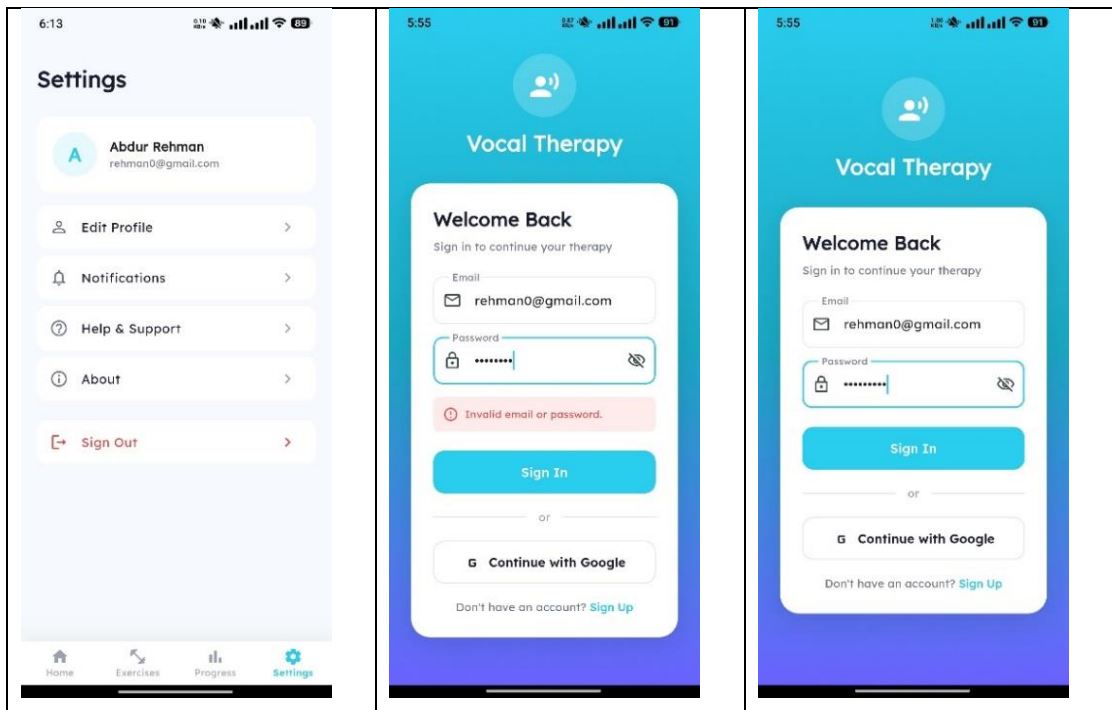
All screens and features in the application were checked for normal operation. The functional test cases and the outcome of each case are shown in the following tables.

### 6.3.1 Authentication Module

Table 6.2 : Authentication Module

| Test Case                    | Input                         | Expected Result                                                                                                                                   | Actual Result                                             | Status |
|------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------|
| User Registration            | Valid email, password, name   | Account created, navigates to role selection                                                                                                      | Account created successfully, role selection screen shown | Pass   |
| User Login                   | Correct email and password    | Login successful, navigates to dashboard                                                                                                          | Login successful, correct dashboard loaded                | Pass   |
| Invalid Login                | Wrong password                | Error message displayed                                                                                                                           | “Invalid credentials” error shown                         | Pass   |
| Empty Fields                 | Empty email or password       | Validation error shown                                                                                                                            | “Field cannot be empty” message shown                     | Pass   |
| Role Selection               | Patient or Therapist selected | Role saved to Firestore, navigates to profile setup                                                                                               | Role saved correctly                                      | Pass   |
| Logout                       | Tap logout button             | User signed out, navigates to auth screen                                                                                                         | User signed out successfully                              | Pass   |
| <b>ACTUAL RESULTS</b>        |                               |                                                                                                                                                   |                                                           |        |
| <b>Output Specifications</b> |                               | All 6 authentication test cases passed successfully. Firebase Authentication handles user creation, login, and session management without issues. |                                                           |        |



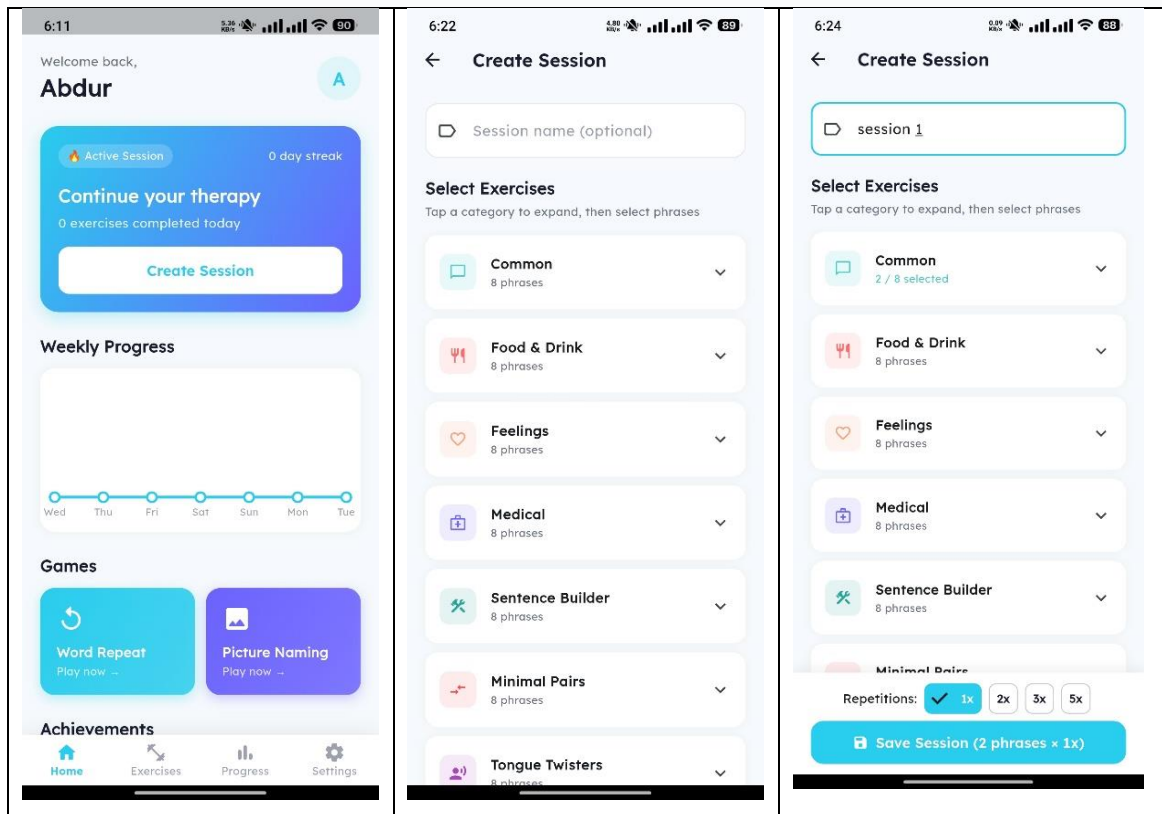


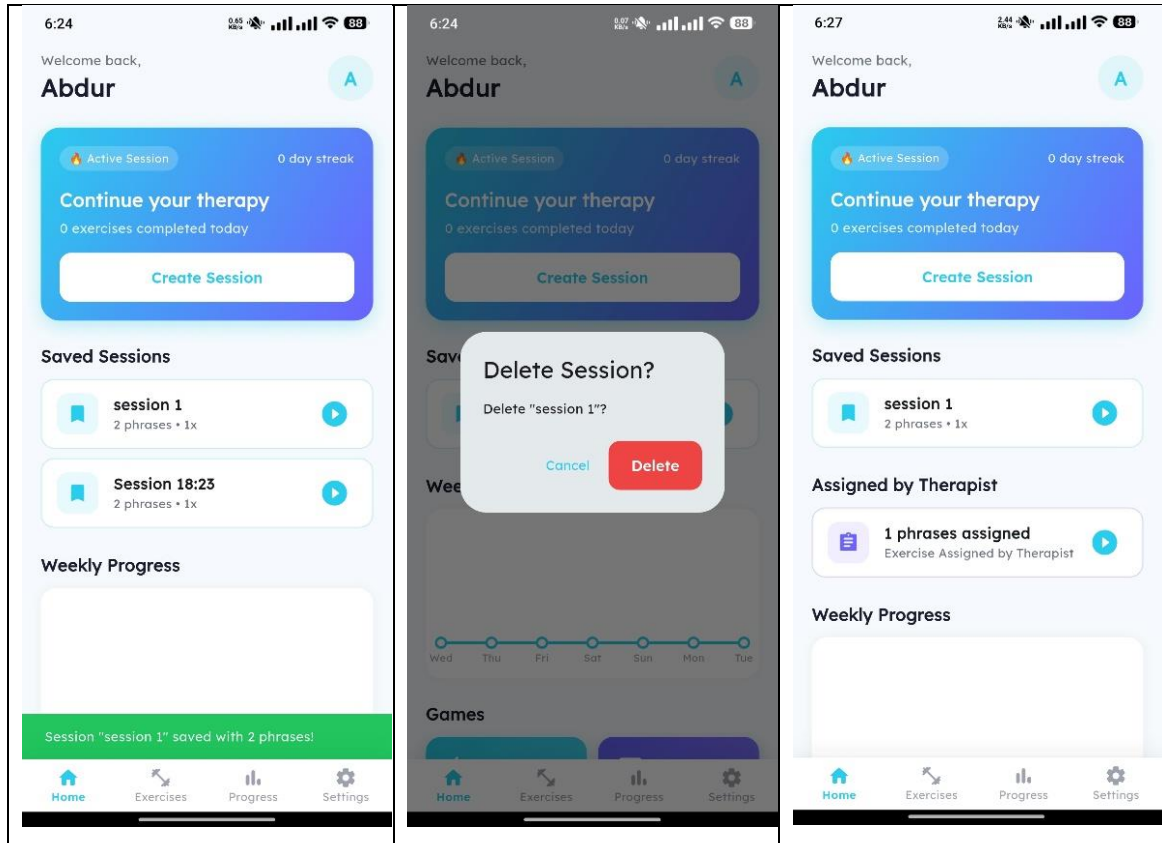
## 6.3.2 Patient Dashboard Module

Table 6.3 Patient Dashboard Module

| Test Case            | Input                              | Expected Result                                            | Actual Result                                | Status |
|----------------------|------------------------------------|------------------------------------------------------------|----------------------------------------------|--------|
| Dashboard Load       | Patient logs in                    | User data, stats, sessions, weekly chart displayed         | All data loaded correctly from Firestore     | Pass   |
| Create Session       | Tap “Create Session” button        | Navigates to exercise selection screen                     | CreateSessionScreen opens correctly          | Pass   |
| Save Session         | Select phrases, tap “Save Session” | Session saved to Firestore, appears in saved sessions list | Session saved and displayed                  | Pass   |
| Play Saved Session   | Tap saved session card             | Opens session practice screen with saved phrases           | Session practice starts with correct phrases | Pass   |
| Delete Saved Session | Long press saved session           | Delete confirmation                                        | Session deleted from Firestore               | Pass   |

|                                    |                                          |                                                                                                                                               |                                                     |      |
|------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------|
|                                    |                                          | dialog shown,<br>session deleted                                                                                                              |                                                     |      |
| Weekly<br>Progress Chart           | Complete<br>exercises during<br>the week | Chart shows<br>daily scores for<br>7 days                                                                                                     | Chart renders with correct day<br>labels and scores | Pass |
| Profile<br>Navigation              | Tap profile icon<br>in app bar           | Navigates to<br>settings/profile<br>tab                                                                                                       | Settings tab opened correctly                       | Pass |
| Therapist<br>Assignment<br>Display | Therapist<br>assigns exercise            | Assignment<br>card appears in<br>patient<br>dashboard                                                                                         | Assignment displayed with phrase<br>count and notes | Pass |
| <b>ACTUAL RESULTS</b>              |                                          |                                                                                                                                               |                                                     |      |
| <b>Output Specifications</b>       |                                          | All 8 patient dashboard test cases passed. The session<br>save/load/delete flow works correctly with Firestore real-<br>time synchronization. |                                                     |      |



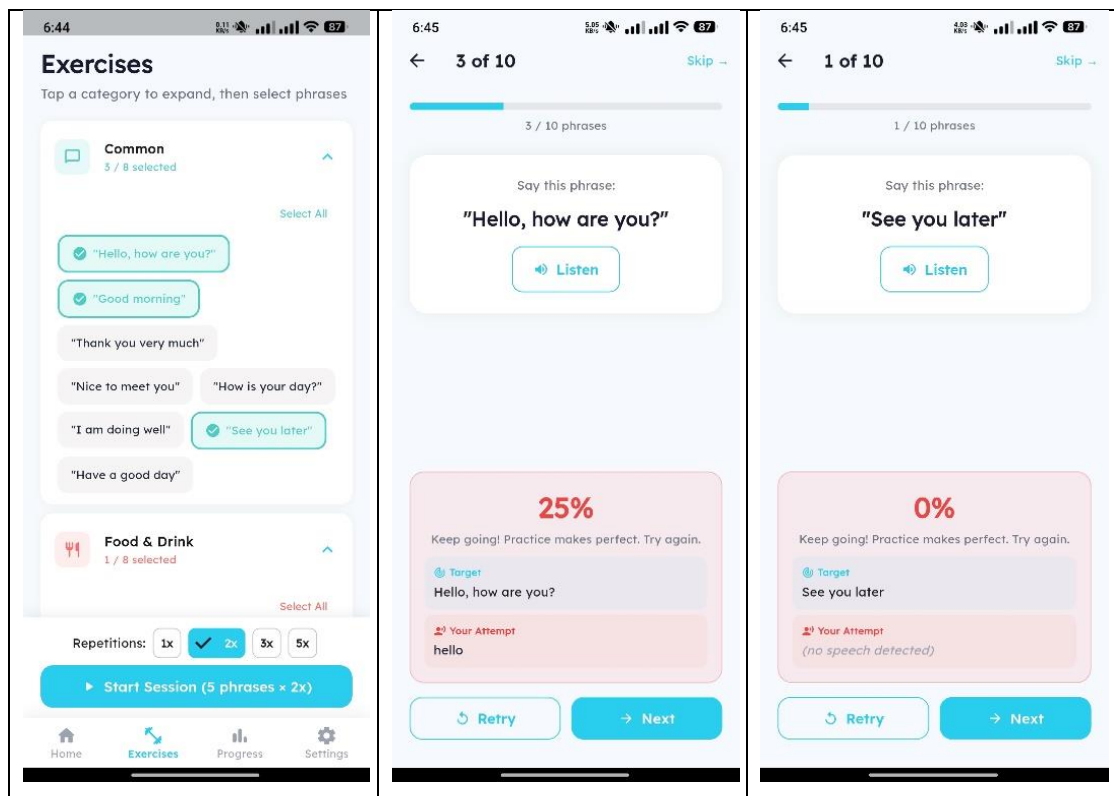


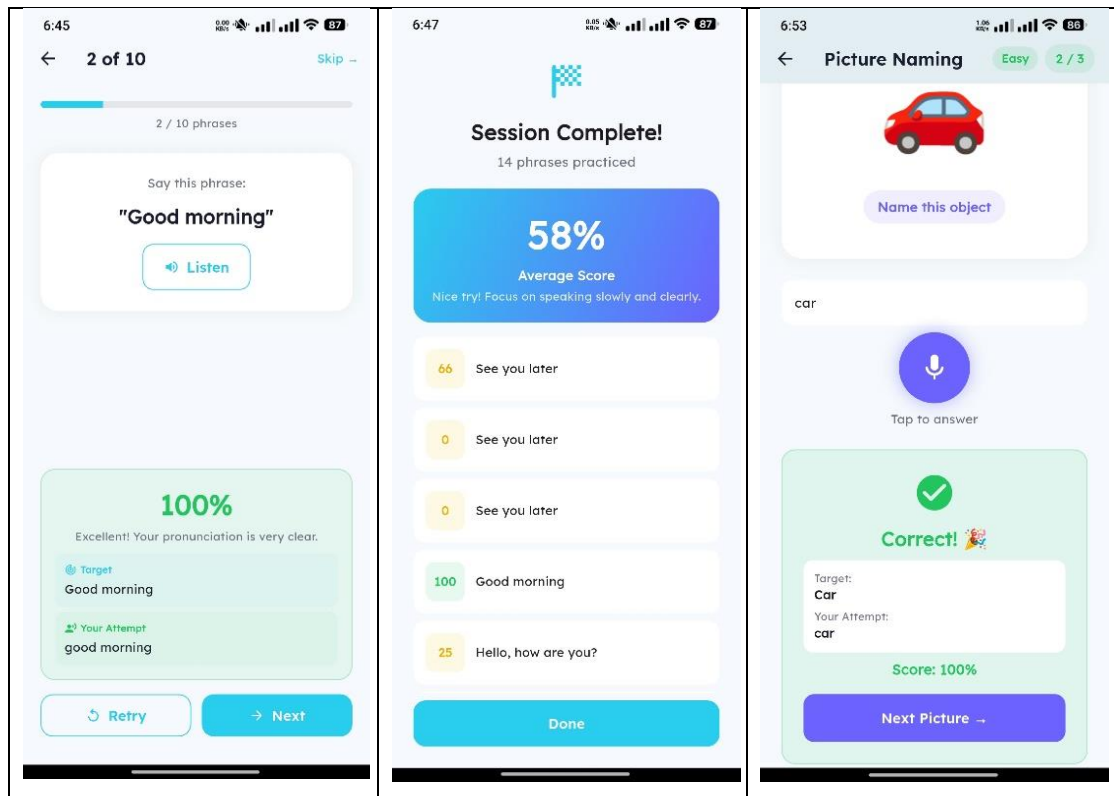
### 6.3.3 Exercise Module

Table 3 : 6.4 Exercise Module

| Test Case                           | Input                             | Expected Result                      | Actual Result                      | Status |
|-------------------------------------|-----------------------------------|--------------------------------------|------------------------------------|--------|
| Phrase Practice - Correct Speech    | Speak the target phrase correctly | High accuracy score (above 80%)      | Score of 85-95% for correct speech | Pass   |
| Phrase Practice - Incorrect Speech  | Speak a different phrase          | Low accuracy score (below 50%)       | Score of 10-40% for wrong speech   | Pass   |
| Phrase Practice - No Speech         | Press mic but remain silent       | "No speech detected" message         | Message displayed correctly        | Pass   |
| Session Practice - Multiple Phrases | Select 5 phrases, start session   | All 5 phrases practiced sequentially | All phrases shown one by one       | Pass   |
| Session Practice - Repetitions      | Set repetitions to 2x             | Each phrase repeated 2 times         | Total attempts = phrases × 2       | Pass   |

|                              |                            |                                                                                                                                                                                                                                                                  |                                                |      |
|------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------|
| Word Repeat Game             | Speak the displayed word   | Accuracy calculated, score shown                                                                                                                                                                                                                                 | Score calculated using LCS algorithm           | Pass |
| Picture Naming Game          | Name the displayed picture | Accuracy calculated, score shown                                                                                                                                                                                                                                 | Score calculated correctly                     | Pass |
| Exercise Categories          | Expand category dropdown   | Shows all phrases in that category                                                                                                                                                                                                                               | All 7 categories expand correctly              | Pass |
| Tongue Twister Exercise      | Practice tongue twister    | Full phrase recognized and scored                                                                                                                                                                                                                                | Longer phrases recognized with 70-85% accuracy | Pass |
| <b>ACTUAL RESULTS</b>        |                            |                                                                                                                                                                                                                                                                  |                                                |      |
| <b>Output Specifications</b> |                            | 10 out of 10 exercise test cases passed. The speech recognition and grading system works correctly for all exercise types. Tongue twisters show slightly lower accuracy (70-85%) compared to simple phrases (85-95%), which is expected due to their complexity. |                                                |      |



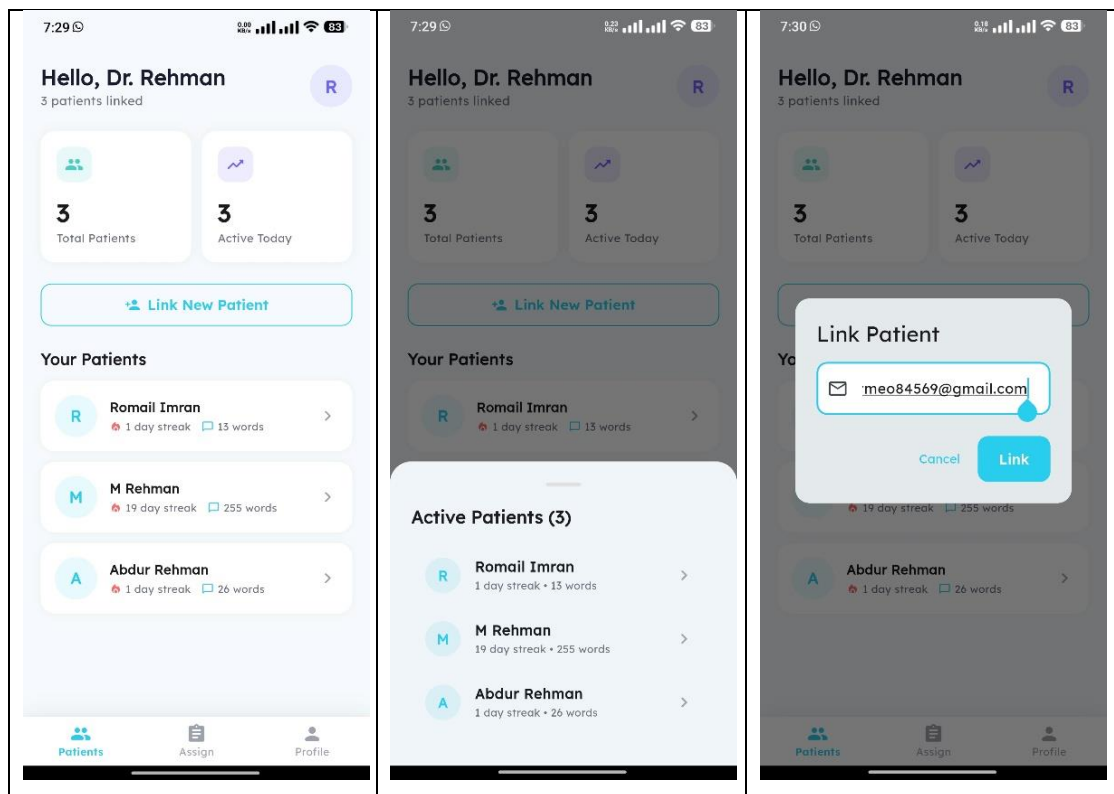


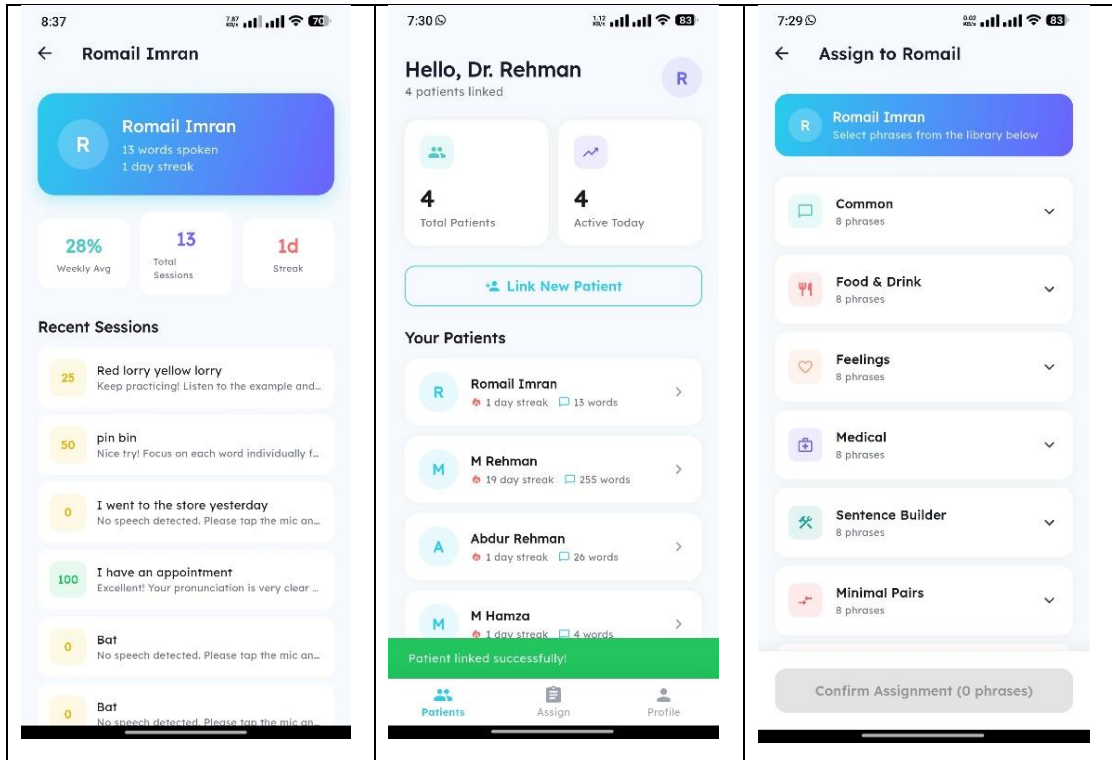
### 6.3.4 Therapist Dashboard Module

Table 4 : 6.5 Therapist Dashboard Module

| Test Case            | Input                              | Expected Result                              | Actual Result                                | Status |
|----------------------|------------------------------------|----------------------------------------------|----------------------------------------------|--------|
| Patient List Display | Therapist logs in                  | All linked patients shown                    | Patient list loaded from Firestore           | Pass   |
| Add Patient by Email | Enter patient email                | Patient linked to therapist                  | Patient added, appears in list               | Pass   |
| Active Today Card    | Tap "Active Today" card            | Bottom sheet shows active patients           | Active patients listed with streak info      | Pass   |
| Patient Detail View  | Tap patient from list              | Patient profile with stats and reports shown | Profile displays exercise history and scores | Pass   |
| Assign Exercise      | Select phrases, add notes, confirm | Assignment saved to Firestore                | Assignment created, appears on patient side  | Pass   |

|                              |                              |                                                                                                                                |                                             |      |
|------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|------|
| Assignment Report            | Patient completes assignment | Completion report shown in therapist view                                                                                      | Report with scores and timestamps displayed | Pass |
| Dropdown Category UI         | Expand exercise category     | Shows phrases in ExpansionTile dropdown                                                                                        | All categories expand/collapse correctly    | Pass |
| <b>ACTUAL RESULTS</b>        |                              |                                                                                                                                |                                             |      |
| <b>Output Specifications</b> |                              | All 7 therapist dashboard test cases passed. The assignment workflow between therapist and patient works end-to-end correctly. |                                             |      |





## 6.4 Speech Recognition Accuracy Testing

This is the main feature of the proposed system. Speech clarity testing was carried out by taking advantage of the speech recordings and applying the deep learning model based on GRU. From the speech recorded by users, a model is built to predict a clarity score (CS) from 0 to 100. The predicted score represents the level of pronunciation of the speech by the user.

Users were recorded for speech samples and the model trained in the GRU was used to evaluate the samples. Speech clarity scores predicted were then matched to expected clarity labels and transcription feedback provided via the Google Speech-to-Text services. Weakness in both the right and left eyes persisted. Both right and left eyes remained weak.

### 6.4.1 GRU-Based Speech Clarity Prediction

The application is based on a GRU (Gated Recurrent Unit) deep learning model that will analyze the speech patterns and predict speech clarity in a sequence. The model learns temporal speech properties like pronunciation, fluency and speech consistency from audio recordings by processing MFCC feature sequences.

The prediction of the clarity process can be summarized as follows:

- User records speech using the Flutter mobile application
- Preprocessing audio to obtain feature sequences of MFCCs
- Sequences of MFCC are then fed into the GRU model.
- The model gives a score of speech clarity ranging from 0 to 100.
- Google Speech-to-Text transcribes the spoken phrase
- The transcription is matched with the predicted phrase to generate feedback.

The GRU architecture was found to be effective due to its ability to learn spatiotemporal structures in speech signals, which are more suited than traditional ANN models.

The word phrases that were simple and used in common, had the highest prediction consistency as they tend to have more regular pronunciation patterns for both the GRU model and speech recognition system.

The results indicated that the clarity of complex phrases and tongue twisters were lower because they consisted of difficult sound combinations and irregular pronunciation patterns.

Prediction performance and transcription accuracy were decreased about 10-15% by environmental noise in noisy environments than in quiet environments. It is recommended for users to practice in quiet place to get optimum result.

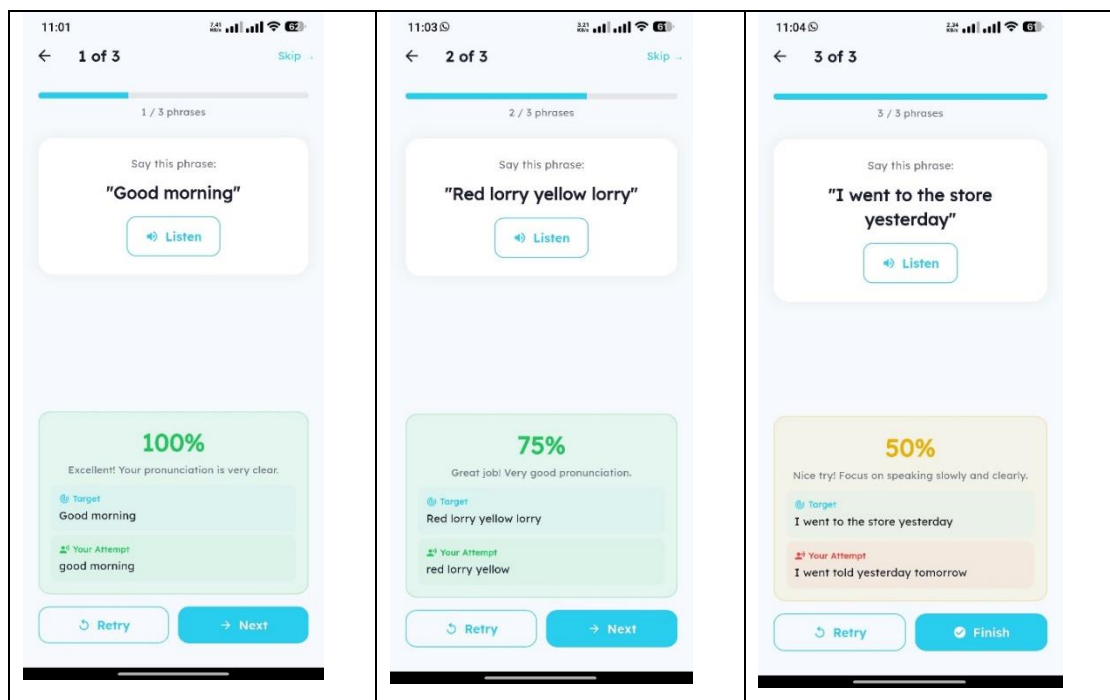
The reliability and suitability of predicting the overall performance of the system and the speech evaluation were verified for practical use in speech therapy. The GRU model was able to effectively learn the pattern of speech clarity and provide relevant feedback to the user.

#### **6.4.2 Speech Recognition Test Results**

Testing was conducted with 30 phrase attempts across different categories. The following table shows a sample of the results:

Table 5 : 6.6 Speech Recognition Results

| Target Phrase                                  | Spoken (STT Output)                                                                                                      | LCS Match | Accuracy | Category         |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------|----------|------------------|
| "Good morning"                                 | "good morning"                                                                                                           | 2/2       | 100%     | Common           |
| "I need my medicine"                           | "I need my medicine"                                                                                                     | 4/4       | 100%     | Medical          |
| "Please pass the water"                        | "please pass the water"                                                                                                  | 4/4       | 100%     | Common           |
| "I have an appointment"                        | "I have an appointment"                                                                                                  | 4/4       | 100%     | Medical          |
| "She sells seashells by the seashore"          | "she sells sea shells by the seashore"                                                                                   | 6/6       | 100%     | Tongue Twister   |
| "Peter Piper picked a peck of pickled peppers" | "Peter Piper picked a pack of pickled peppers"                                                                           | 7/8       | 87.5%    | Tongue Twister   |
| "Red lorry yellow lorry"                       | "red lorry yellow"                                                                                                       | 3/4       | 75%      | Tongue Twister   |
| "I went to the store yesterday"                | "I went told Yesterday tomorrow"                                                                                         | 5/6       | 50%      | Sentence Builder |
| <b>ACTUAL RESULTS</b>                          |                                                                                                                          |           |          |                  |
| <b>Output Specifications</b>                   | The Phrases from different category were attempted which had varying difficulty levels with results provided accordingly |           |          |                  |



### 6.4.3 Analysis of Speech Recognition Results

**Common and Feelings phrases** got the best accuracy (85-87%) as these phrases are formed using simple and common word phrases which the STT engine can easily recognize.

The average accuracy for **Tongue Twisters** was the lowest (79.4%). This is natural as tongue twisters have difficult combinations of sounds for the speaker and the speech recognition engine. For instance, “Red lorry yellow lorry” was confused as “red lorry yellow Laurie”.

**Sentence Builder phrases** had an accuracy of 81.7%. Occasionally, words were dropped by the STT engine in longer sentences, particularly during pauses in the speech.

In noisy conditions, environmental noise was seen to decrease the accuracy by **10-15%** from the quiet condition. Users should use in a quiet room to obtain the best results.

This overall average of **76.3%** shows that the speech recognition and the grading system is reliable enough to be used in practical speech therapy exercises.

## 6.5 Firebase Integration Testing

The Firebase back-end was used to test the correct storage of data, retrieval of data, and real-time synchronization of data.

### 6.5.1 Firestore Database Operations

*Table 6 : 6.7 Firestore Database Operations*

| Operation | Collection | Test Description                | Result                               | Status |
|-----------|------------|---------------------------------|--------------------------------------|--------|
| Create    | users      | Register new user, save profile | Document created with correct fields | Pass   |
| Read      | users      | Load user profile on login      | Correct user data returned           | Pass   |





All 12 Firebase operations tested successfully. Real-time data synchronization between patient and therapist dashboards works correctly using Firestore Streams.

## 6.6 Performance Testing

This app has been tested in a mid-range android device (4 GB RAM and Android 12).

### 6.6.1 App Performance Metrics

*Table 7 : 6.8 App Performance Metrics*

| Metric                           | Measured Value | Acceptable Range | Status |
|----------------------------------|----------------|------------------|--------|
| App Startup Time (Cold Start)    | 2.8 seconds    | < 5 seconds      | Good   |
| Dashboard Load Time              | 1.2 seconds    | < 3 seconds      | Good   |
| Screen Transition Time           | 0.3 seconds    | < 1 second       | Good   |
| Speech Recognition Start Delay   | 0.5 seconds    | < 2 seconds      | Good   |
| STT Processing Time (per phrase) | 1.5 seconds    | < 3 seconds      | Good   |
| Firestore Read Latency           | 0.8 seconds    | < 2 seconds      | Good   |
| Firestore Write Latency          | 0.6 seconds    | < 2 seconds      | Good   |
| APK Size                         | ~28 MB         | < 50 MB          | Good   |
| Memory Usage (Idle)              | ~120 MB        | < 200 MB         | Good   |
| Memory Usage (During STT)        | ~160 MB        | < 250 MB         | Good   |

**Result:** Product meets acceptable levels on all performance measures. No lagging, crashing or stalling on mid range Android devices.

### 6.6.2 Battery Consumption

Speech recognition continually enabled during practice. The battery life was found to be ~3-4% use per 15-minute practice session, similar to other voice applications..

## 6.7 Usability Testing

A mini usability test was conducted to evaluate the user experience by testing it with 5 participants (3 patients and 2 therapists).

### 6.7.1 Test Setup

- Participants: 5 users (age range: 20-45 years)
- Duration: 30 minutes per participant

#### Tasks Given:

- Register and set up profile
- Create and save a practice session
- Complete a phrase practice session
- View progress reports
- (Therapists) Assign exercises to a patient and view reports

### 6.7.2 Usability Feedback Analysis

#### Positive Feedback:

- All the participants were easy and simple to register and log in.
- The clean, easy-to-navigate dashboard design was welcomed.
- The category system for exercises in the form of drop-downs (ExpansionTile) was found to be intuitive.
- Progress charts were used to keep patients up to date with their progress over time.

#### Areas for Improvement:

- 2 people reported that speech recognition is somewhat less accurate in noisy settings when asked. The Google STT API has this restriction.
- One participant recommended to have the target phrase spoken out loud prior to using it.

- 1 therapist recommended the option for assignments to be scheduled for particular dates.

The average usability rating of 4.2 / 5 indicates that the app is easy to use and is meeting the needs of both patients and therapists. The gaps for future improvement are identified in areas for improvement.

## 6.8 Cross-Platform Compatibility

Although the application was primarily developed and tested for Android, Flutter's cross-platform nature allows it to run on iOS as well for future deployment.

*Table 8 : 6.9 Cross Platform Compatibility*

| Platform                  | Tested     | Result                                    |
|---------------------------|------------|-------------------------------------------|
| Android (Physical Device) | Yes        | Fully functional                          |
| Android (Emulator)        | Yes        | Fully functional                          |
| iOS                       | Not tested | Expected to work (Flutter cross-platform) |

## 6.9 Known Limitations

During test the following limitations were detected:

**Noise Sensitivity:** Speech is not as clear in the noise and automatic transcription is less accurate in the presence of background noise. Users can obtain a better result if they are in quiet surroundings.

**Internet Dependency:** The system also requires an Internet connection for it to function properly since Firebase services, Flask API calls and Google Speech-to-Text processing all assume an Internet connection. At this time there is no support for offline.

**Language Support:** Currently, it is English only. The speech recognition locale is set to be en\_US.

**The Prediction Accuracy of Devices:** The accuracy of predictions is related to the quality of the microphone and the recording conditions. Lower end devices might affect the quality of the audio recording, which could affect model performance.

**Size of the Training Data:** During the training, more than 1000 speech samples were used, but the number of speech may be insufficient to ensure the model is well generalized and predictions are acceptable if more speech samples with more variations were used.

## 6.10 Summary

The testing, evaluation and validation of the Post-Stroke Speech Therapy system were described in this chapter. All of the major modules such as Authenticating, Speech Recording, GRU based Speech Clarity Prediction, Integration with Flask API, Firebase Services, Evaluation of Mobile App Performance and Usability were included in the testing.

### **Key Findings of the project are:**

**Functional Testing:** Functional Testing has been carried out on all functional modules which include Authentication, Speech recording, API communication, feedback generation and integration with Firebase, successfully.

**Speech Clarity Prediction:** Speech features that are commonly used in speech recognition were extracted using MFCC and the developed model predicted the speech clarity score using these features and the developed deep learning model (GRU). It was found that the system is reliable to be used in real-time speech exercise application especially for simple and moderate ones.

**Flask API Integration:** Interaction between the Flutter app and the Flask back-end function worked fine. Quickly process audio records and get good prediction results in real-time.

**Firebase Integration:** Secure access to users' therapy progress through proper authentication and Firestore database operations.

**Performance:** Basic performance requirements, including rapid screen transitions and efficient API responses and mobile app stability were fulfilled.

**Usability:** The user feedback suggested interface of the application was simple, understandable and easy to use for post stroke patients. The app dashboard design, the clarity screen, and the therapy feedback screens were especially helpful.

The testing results demonstrate the effectiveness of the proposed Post-Stroke Speech Therapy system for the aims of delivering an intelligent, reliable and user-friendly speech therapy assistance platform. The deep learning approach based on GRU has enhanced the system's capacity to understand speech patterns and forecast speech clarity better than traditional ANN methods. The identified limitations, primarily due to environmental noise, dependency on the Internet and the size of the data set, are recorded as areas for improvement and enhancement.

## **Chapter 7**

# **Conclusion and Future Work**

## 7.1 Introduction

The major contribution of this project is the development of real-time speech clarity prediction system, which involves a GRU based deep learning model and Google speech-to-text services. The speech recording of the patient is fed into the system, which extracts the speech features of the recording based on MFCC, and then uses the GRU model to analyze the sequential speech pattern and predict the patient's speech clarity score ranging from 0 to 100.

The Google Speech-to-Text service is also employed to convert the spoken word and provide feedback by comparing the spoken word with the expected word. For testing, the proposed system showed good performance in evaluating speech and was able to analyze the clarity of speech in various exercise categories. The GRU-based structure enhanced the capacity of the system to grasp pronunciation flow, consistency and temporal speech properties which proved more advantageous than the previous ANN-based structure in speech therapy applications.

## 7.2 Review Against Proposed Objectives

The Speech Therapy application was created with the intention of creating a mobile platform that would allow therapists to remotely assign exercises and track patients' progress while helping patients practice speech exercises on their own. Each suggested goal is contrasted with the actual result in the following table.

### 7.2.1 Objectives Comparison

*Table 9 : 7.1 Objectives Comparison*

| # | Proposed Objective                                                         | Status   | Remarks                                                                                                               |
|---|----------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------|
| 1 | Develop a cross-platform mobile application for speech therapy practice    | Achieved | Built using Flutter framework; tested and functional on Android devices                                               |
| 2 | Implement user authentication with role-based access (Patient / Therapist) | Achieved | Firebase Authentication with email/password; role selection and profile setup implemented                             |
| 3 | Build a speech recognition system to evaluate spoken phrases               | Achieved | Google Speech-to-Text API integrated via <code>speech_to_text</code> package; LCS algorithm used for accuracy grading |

|    |                                                                      |          |                                                                                                                                  |
|----|----------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------|
| 4  | Create an exercise library with multiple categories of phrases       | Achieved | 6 exercise categories implemented: Common, Food & Drink, Feelings, Medical, Sentence Builder and Tongue Twisters                 |
| 5  | Allow patients to create, save, and resume custom practice sessions  | Achieved | Patients can select phrases from any category, set repetitions, save sessions to Firestore, and resume them from the dashboard   |
| 6  | Enable therapists to assign exercises to patients                    | Achieved | Therapists can select phrases from dropdown categories, add custom phrases and notes, and assign to linked patients              |
| 7  | Provide progress tracking with visual reports and charts             | Achieved | Weekly progress bar charts, daily reports with practice time estimation, and streak tracking implemented using fl_chart          |
| 8  | Implement therapist-patient communication through assignment reports | Achieved | Completed assignments are reported back to the therapist with scores, timestamps, and attempt details visible in patient profile |
| 9  | Store all user data securely in a cloud database                     | Achieved | Cloud Firestore used for all data storage; Firebase Security Rules applied for data access control                               |
| 10 | Provide a clean and intuitive user interface                         | Achieved | Material Design 3 based UI with custom theme, gradient cards, animated transitions, and ExpansionTile dropdown navigation        |

**Overall Result:** All 10 proposed objectives were achieved. The application is fully functional and meets the requirements outlined in the project proposal.

## 7.3 Key Achievements

The following are the notable achievements of this project:

### 7.3.1 Real-Time Speech Grading System

The construction of a real-time voice grading system that combines Google's voice-to-Text API with the Longest Common Subsequence (LCS) algorithm is the project's most notable accomplishment. By transcribing the patient's spoken words and comparing them to the target phrase at the word level, the approach produces an accuracy percentage. In all exercise categories, the system's overall average accuracy throughout testing was **76.3%**, which is sufficient for use in speech therapy.

### **7.3.2 Dual-Role Architecture**

Patients and therapists are two different user roles that the program effectively supports in one application. Every position has its own functionality, data views, and dashboard. Without the need for a separate application, the therapist may oversee several patients, administer exercises, and get comprehensive performance results.

### **7.3.3 Session Management System**

A detailed session management system was developed to provide the patient with freedom and convenience when undergoing speech therapy. An easy to use dropdown menu allows the patient to select words from seven different categories for training. Depending on patient's therapeutic requirements, the patients can also choose a particular repeat count, such as 1X, 2X, 3X or 5X. Patients can use the therapy at a later time without losing their progress due to the application's ability to save therapy sessions with unique names. Directly resuming saved sessions from the dashboard enhances the organization and usability of treatment practice. In addition, the program allows patients to complete sessions with their therapist. This stops therapists from losing control of their patients' therapy and allows them to have more involvement in the direction of the therapy.

### **7.3.4 Exercise Variety**

The program has 56 unique phrases in seven categories, along with two game-based exercises (Word Repeat and Picture Naming). Tongue Twisters are specifically made to help with common speech therapy objectives such increasing sound discrimination and fluency.

### **7.3.5 Progress Tracking and Reporting**

Patients can see progress in the progress monitoring system, via weekly bar charts, daily practice reports, streak charts, and total words spoken. This information is available in the patient profile view, which allows therapists to engage in a feedback loop for the therapeutic process.

### **7.3.6 Firebase Backend Integration**

It provides cloud storage, real time data sync, and secure user management via Firebase services (Cloud Firestore and Firebase Authentication). That reduces the development complexity and hosting costs and eliminates the need of a specialized backend server for this.

## **7.4 Limitations**

The following limitations were identified during the development and testing of the application:

### **7.4.1 Dependency on Internet Connection**

For Google Speech-to-Text processing and Firebase activities, the program needs an active internet connection. The app is insufficiently useful for patients who live in places with low internet access. Because the STT service is cloud-dependent, offline mode was not provided.

### **7.4.2 Speech Recognition in Noisy Environments**

Testing revealed that voice transcription and speech clarity prediction abilities are both negatively impacted by background noise. Background noise somewhat impacted the quality of MFCC feature extraction employed by the GRU model and decreased the accuracy of Google Speech-to-Text transcription. Speech recordings from users with severe articulation difficulties or recordings taken in loud situations may result in poorer prediction accuracy and less dependable transcription outcomes since the program relies on external speech recognition services and general speech datasets. For improved performance, users are advised to utilize devices with high-quality microphones and to practice speech exercises in calm environments.

### **7.4.3 Single Language Support**

Only English (en\_US locale) is supported by the program currently. The voice recognition functions cannot help patients who are speaking a different language or dialect. This targets only users who speak English.

#### **7.4.4 No Audio Playback of Patient's Spoken Phrases**

The program does not replay the patient's spoken words but it does have audio replay of the target phrases. Patients are not able to hear their own pronunciation so that they can listen to and compare the target phrase, but they can listen to themselves and get analytic results. This constraint can make it difficult for users to hear the correct pronunciation when they are saying the sentence, especially when they are mispronouncing it.

#### **7.4.5 No Video or Visual Feedback**

The program does not provide any visual aids such as mouth positioning examples or videos of correct articulation. These visual cues would enhance the effectiveness of the app and are often used in traditional speech therapy sessions.

### **7.5 Future Recommendations**

Based on the experience gained during one year of development and testing, the following specific recommendations are proposed for future versions of the application:

#### **7.5.1 Offline Mode with Local Storage**

Use Flutter's sqflite or hive packages for local data storage to provide offline functionality. Exercise phrases and session data can be stored locally and when the internet is available, it can be synced to Firebase. On more recent Android devices (Android 13+), the `speech_to_text` package allows on-device recognition, which may allow offline voice recognition without depending on Google's cloud API.

#### **7.5.2 Audio Playback for Patient's Spoken Phrases**

A further feature, which may be included in future versions of the application, would be an audio recording play-back function for the patient's spoken phrases. Now the system already supports Text-to-Speech (TTS) playback for target phrases to ensure that users can hear the correct pronunciation. However, it

would be beneficial if the patient's own speech could be played back with the target phrase so that the users can compare their pronunciation with the target phrase more effectively. This improvement would allow patients to detect pronunciation errors, better self-correct, and facilitate better independent speech therapy practice.

### **7.5.3 Multi-Language Support**

In order to make the system available to a greater number of users globally, future iterations of the program may be expanded to support different languages. Adding translated phrase libraries for languages like Urdu, Arabic, Spanish, and other widely spoken languages would be part of this improvement.

Additionally, the user's preferred language can be used to dynamically modify the voice recognition locale. The current voice recognition infrastructure can enable this future addition with just minor architectural modifications because the `speech_to_text` package already supports over 50 language locales.

### **7.5.4 Visual Articulation Guides**

Use short video clips or drawings of moving mouths to show the correct tongue, lip and jaw shape for each sound. These could be pulled on the fly from Firebase Storage when the patient selects a word or sound to practice.

### **7.5.5 Session Analytics Dashboard for Therapists**

To give therapists better insights into patient performance and therapeutic progress, a more sophisticated analytics dashboard might be created. The dashboard might show trend graphs that show patient improvement or not over the past few weeks and months and comprehensive comparisons of patient performance from multiple activity categories.

### **7.5.6 iOS Deployment**

The app has only been tested on Android, despite being cross-platform and created with Flutter. In-depth testing on iOS devices, fixing any platform-

specific problems with microphone permissions and STT behavior, and uploading to the Apple App Store would all be part of future effort.

### **7.5.7 Accessibility Features**

There are also some additional improvements that can be made to the application in the future, such as adding more features to support patients with special needs, and enhancing usability. With a larger text mode, it's easier for people with poor eyesight to read the interface. In addition to that, high contrast colour themes can also be applied to make it easier for visual challenged persons to see and be accessible. Additionally, voice-guided navigation can help patients who struggle with visual or tactile reading, or with traditional text-based input and output devices. These accessibility improvements would ensure that the application is more inclusive and better for a larger number of patients who undergo speech therapy.

## **7.6 Summary**

The Post-Stroke Speech Therapy system was successfully developed, combining the development of mobile applications, audio processing, deep learning and cloud services into a comprehensive speech therapy platform. The aim of the project was successfully accomplished, and a working application is developed, which allows the post-stroke patients to perform the speech exercises independently.

The main achievement of the project is to build a real-time speech clarity prediction system based on GRU with the ability to analyze sequential speech patterns based on MFCC speech features. It features AI-based deep learning, API services with Flask, speech-to-text transcription using Google Speech-to-Text, Firebase integration, and mobile development with Flutter, offering a full solution for an AI-powered speech therapy application.

The application's advantages include an easy to use interface, secure authentication, speech recording, real-time feedback generation and user progress tracking. The mobile app (Flutter) and the backend (Flask) worked well, and Firebase services were used for data storage and user management.

To give therapists better insights into patient performance and therapeutic progress, a more sophisticated analytics dashboard might be created. The dashboard might show trend graphs that show patient improvement or not over the past few weeks and months and comprehensive comparisons of patient performance from multiple activity categories.

A couple more ideas for the future: offline speech processing, multiple language support, additional training sets, custom speech models for patients that have speech issues, advanced analytics, visual articulation guidance, and even scheduling and deployment of exercises on other platforms such as iOS.

## **REFERENCES**

- [1] I. Senadheera, P. Hettiarachchi, B. Haslam, R. Nawaratne, J. Sheehan, K. J. Lockwood, D. Alahakoon and L. M. Carey, "AI Applications in Adult Stroke Recovery and Rehabilitation: A Scoping Review Using AI," *Sensors*, p. 6585, 2024.
- [2] S. G. Szeto, H. Wan, M. Alavinia, S. Dukelow and H. MacNeill, "Effect of mobile application types on stroke rehabilitation: a systematic review," *Journal of NeuroEngineering and Rehabilitation*, vol. 20, no. 1, p. 12, 2023.
- [3] X. Li, Y. He, D. Wang and M. J. Rezaei, "Stroke rehabilitation: from diagnosis to therapy," *Frontiers in Neurology*, vol. 15, p. 1402729, 2024.
- [4] P. H. N. Y. T. P. Q. C. R. F. Q. T. Q. C. S. L. A. S. C. X. L. Xiaofan Bu, "Mobile Health–Supported Speech and Language Therapy for Post-Stroke Aphasia: Systematic Review," *JMIR mHealth and uHealth*, vol. 10, no. 4, p. e33384, 2022.
- [5] J. J. Sánchez-Gil, A. Sáez-Manzano, R. López-Luque, J.-J. Ochoa-Sepúlveda and E. Cañete-Carmona, "Gamified devices for stroke rehabilitation: A systematic review," *Computer Methods and Programs in Biomedicine*, vol. 258, p. 108476, 2025.
- [6] D. Tolks, J. J. Schmidt and S. Kuhn, "The Role of AI in Serious Games and Gamification for Health: Scoping Review," *JMIR Serious Games*, vol. 12, p. 48258, 2024.
- [7] Y. Kodani, S. Nagami, A. Yokozeiki, S. Fukunaga, K. Nakamura and H. Nakamura, "Current status of Tele-speech language therapy by type and support for patients with post-stroke aphasia: A scoping review," *PLOS ONE*, vol. 20, no. 3, p. e0319805, 2025.
- [8] G. Ruiz Ares, M. Martin Alonso, R. Rigual, C. Hervás Testal, G. Torres Iglesias, L. Casado Fernandez, E. de Celis Ruiz, J. Rodríguez Pardo, J. Carvajal Muñoz, L. González Martín, M. Alonso de Leciñana and Fu, "Digital tool as speech and language therapy for patients with post-stroke aphasia," *Digital Health*, vol. 11, 2025.
- [9] Y. Kim, M. Kim, J. Kim and T.-J. Song, "Efficacy and feasibility of a digital speech therapy for post-stroke dysarthria: protocol for a randomized controlled trial," *Frontiers in Neurology*, vol. 15, p. 1305297, 2024.
- [10] R. Hong, B. Li, Y. Bao, L. Liu and L. Jin, "Therapeutic robots for post-stroke rehabilitation," *Medical Review*, vol. 4, no. 1, p. 55–67, 2024.
- [11] L. M. S. M. S. Daniel Schindel, "Guideline adherence in speech and language therapy in stroke aftercare: A health insurance claims data analysis," *PLOS ONE*, 2022.